

OPTIMAL FEEDBACK LAW RECOVERY BY GRADIENT-AUGMENTED NEURAL NETWORK WITH NONCONVEX REGULARIZATION ON THE UNBOUNDED PARAMETER SPACE

ABSTRACT. We recover optimal feedback laws for nonlinear control problems by approximating Hamilton–Jacobi–Bellman value functions with sparse shallow neural networks. Value and gradient are generated from the open-loop problem by the Pontryagin maximum principle, and the network is trained with a gradient-augmented H^1 fidelity.

For nonhomogeneous activation functions on the unbounded parameter space $\Omega = \mathbb{R}^{d+1}$, fix $p > 0$, set $w_p(\omega) = 1 + |\omega|^p$ and $\mu_p = w_p \mu$, and normalize the ridge atom K as $K_p = K/w_p$. We study the normalized objective

$$J(\mu) = L(\mu) + \alpha \Phi_\phi(\mu_p),$$

where the concave scalar penalty has finite positive right slope at zero. Normalization makes the dictionary vanish at infinity under a polynomial growth gap. This yields existence by normalized weak-* compactness. If the penalty is strongly concave on every bounded interval, every local minimizer has finite support and satisfies a covering-number bound, while the global minimum is attained by a finite network. Maximization of the weighted Gâteaux derivative in the a posteriori bound at each insertion, followed by a subsequent outer weights correction, achieves a best-iterate rate of order $n^{-1/2}$.

For positively k -homogeneous activations, we give an equivalent measure problem on the unit sphere with the fractional outer-coefficient penalty $|c|^{2/(k+1)}$. Every local minimizer in total variation has finite support, while every global minimizer satisfies an explicit a priori width bound. The resulting algorithms are studied on the Van der Pol oscillator and the pendulum swing-up problem, whose value function has a gradient discontinuity across a switching set.

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1. INTRODUCTION

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$(P) \quad \begin{aligned} & \inf_{u \in L^2(t_0, T; \mathbb{R}^{m_u})} \mathcal{C}(u; t_0, x) := \int_{t_0}^T (h(y(t)) + \eta |u(t)|^2) dt, \quad \eta > 0, \\ & \text{s.t.} \\ & \dot{y} = f(y) + g(y)u, \\ & y(t_0) = x, \end{aligned}$$

We assume $y(t) \in \mathbb{R}^d$, $u(t) \in \mathbb{R}^{m_u}$, $h \in C^1(\mathbb{R}^d)$, $f \in C^1(\mathbb{R}^d; \mathbb{R}^d)$, and $g \in C^1(\mathbb{R}^d; \mathbb{R}^{d \times m_u})$. The control-theoretic results below are used under the following standing well-posedness assumptions: for every admissible initial pair and every $u \in L^2(t_0, T; \mathbb{R}^{m_u})$, the state equation has a unique absolutely continuous solution; the value is finite and the infimum is attained; every minimizer satisfies the Pontryagin maximum principle; and the dynamic programming principle and comparison hypotheses hold, so that the continuous value function is characterized as the viscosity solution of the HJB equation.

One approach computes an open-loop state-costate-control path for a fixed initial condition x . The Pontryagin maximum principle (PMP) yields a coupled forward-backward two-point boundary value problem:

$$(TPBVP) \quad \begin{cases} \frac{d}{dt} y^* = f(y^*) + g(y^*)u^*, \\ y^*(t_0) = x, \\ -\frac{d}{dt} p^* = \partial_y \left(f(y^*) + g(y^*)u^* \right)^\top p^* + \partial_y h(y^*), \\ p^*(T) = 0. \end{cases}$$

The stationarity condition with respect to the control gives

$$u^*(t) = -\frac{1}{2\eta} g(y^*(t))^\top p^*(t) \quad \text{for almost every } t \in (t_0, T).$$

The open-loop control is not robust with respect to disturbances of the state [1]: it is computed for a single initial condition, and deviations from the optimal path are not corrected. In addition, numerical methods for the two-point boundary value problem are sensitive to the initial guess and may fail to converge. The dynamic programming principle instead introduces the value function

$$V(t, x) := \inf_u \mathcal{C}(u; t, x)$$

which satisfies the Hamilton–Jacobi–Bellman (HJB) equation in the viscosity sense:

$$(HJB) \quad \begin{cases} \partial_t V - \frac{1}{4\eta} \nabla V(t, x)^\top g(x) g(x)^\top \nabla V(t, x) + \nabla V(t, x)^\top f(x) + h(x) = 0, \\ V(T, x) = 0. \end{cases}$$

At points where V is differentiable, the equation holds classically and Hamiltonian minimization gives the feedback $u^*(t, x) = -(2\eta)^{-1} g(x)^\top \nabla V(t, x)$. The value function encodes the optimal cost from every state. The feedback law synthesized from it corrects deviations from a nominal path.

For nonlinear control problems one usually seeks a feedback law, which requires solving the HJB equation. Computing this function is notoriously difficult because of the curse of dimensionality. Alongside classical methods, data-driven methods have therefore become increasingly common. One line of work, introduced in [18], generates the training data with (TPBVP) and approximates the optimal value function from it. The optimal feedback law is then synthesized from the fitted

value function after validation. Its samples are causality-free: generating one does not require approximate values at neighbouring grid points. Moreover, Azmi, Kalise and Kunisch [1] enrich the dataset with adjoint information: a state-adjoint-control triple is generated by solving the open-loop problem, and at differentiability points the adjoint supplies the value gradient. The value function is then approximated by hyperbolic-cross basis functions with an ℓ^1 penalty, and including the adjoint data targets an H^1 approximation. Neural-network approximations form another growing line of work [16, 23]. Other prior information, such as semiconcavity of the value function, can also be incorporated. Alternatively, one can use an HJB- or physics-informed loss [9, 17, 22, 24, 15].

For $\omega = (\omega_n)_{n=1}^N$, $\omega_n = (a_n, b_n)$, and $c = (c_n)_{n=1}^N$, we use the two-layer network

$$(1) \quad \mathcal{N}_{\omega,c}(x) := \sum_{n=1}^N c_n \rho(a_n \cdot x + b_n) \in \mathbb{R},$$

where ρ is a sufficiently smooth activation function and N the width of the network. We associate it with the finite signed measure $\mu_{\omega,c} := \sum_{n=1}^N c_n \delta_{\omega_n}$. Its gradient is

$$\nabla_x \mathcal{N}_{\omega,c}(x) = \sum_{n=1}^N c_n \nabla_x \rho(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

For the finite-horizon problem, we write $V(x) := V(0, x)$. The infinite-horizon pendulum value considered below is stationary, so it has no suppressed time argument. The training data are generated by solving (TPBVP). For the trajectory issued from x^m , let

$$g^m := p^{*,m}(0)$$

denote the costate. At every sample where V is differentiable, sensitivity relations give $g^m = \nabla V(x^m)$ [8, 6, 7]. At a switching point, competing optimal paths may supply different costate labels; no classical gradient is asserted there. Thus the data consist of triples $\{x^m, V(x^m), g^m\}_{m=1}^M$. We have the minimization problem

$$\min_{N,\omega,c} l^M(\mu_{\omega,c}), \quad l^M(\mu_{\omega,c}) := \frac{1}{2M} \sum_{m=1}^M (|V(x^m) - \mathcal{N}_{\omega,c}(x^m)|^2 + |g^m - \nabla \mathcal{N}_{\omega,c}(x^m)|^2).$$

Fix a bounded Lipschitz training domain $D \subset \mathbb{R}^d$. Section 3 introduces the corresponding population fidelity L with respect to Lebesgue measure on D .

Another focus of this study is the sparsity of the network. The number of neurons N is itself an optimization variable. Although width alone does not generally explain a model's inductive bias [19], sparsity is crucial for computational efficiency in many problems. Furthermore, sparsity is necessary for certain applications, such as knowledge distillation and deconvolution of point-like objects.

A traditional sparse shallow network uses the ReLU activation and an ℓ^1 penalty on the outer coefficients [4, 2]. The convex penalty may retain redundant neurons whose inner parameters cluster. Pieper and Petrosyan [20] showed that convex penalties cannot effectively eliminate this redundancy when neurons cluster in a small neighborhood and introduced a concave nonconvex functional for this purpose. However this formulation does not extend to the gradient fitting since the formulation is restricted to the ReLU setting: the second-order weak derivative of ReLU does not exist as a function while it is a distribution, and we cannot use it to approximate high-order derivatives. We need to extend to the smooth activation functions.

The difficulty is, such extension requires a treatment of the unbounded inner-parameter space for gradient training. 1. For positively k -homogeneous activations, radial scaling can be absorbed into the outer coefficient, and the network has an equivalent representation with inner parameters on the unit sphere. However, the induced nonconvex penalty admits infinite slope at 0 and the usual functional tools to prove the existence and optimality are not applicable. 2. For

nonhomogeneous activations such as tanh, softplus, and the Gaussian, radial scaling changes the atom and cannot be removed by normalization. Without normalization, sequences with $|\omega| \rightarrow \infty$ may retain a nonzero contribution to the represented function, as shown in Section 3.3. We address those difficulties in each case.

Contributions. The contributions are the following.

- (1) *Nonhomogeneous activations on the unbounded parameter domain.* We adopt the objective function with gradient information from [1] and the nonconvex penalty from [20]. Like in [20], we require the concave scalar penalties to have finite positive right slope. However, unlike the original paper which only considers the ReLU case, we allow the activation functions to be nonhomogeneous defined on the unbounded domain. We prove a weak representation of the formulation and derive weak-* lower semicontinuity (Theorem 3.9 and Corollary 3.10). In Theorem 3.13 with the normalized atom K_p , we extend the existence result of a global minimizer from the value loss with ReLU activation function to the gradient-augmented loss with any continuous activations on $\Omega = \mathbb{R}^{d+1}$ of implied polynomial growth condition (Theorem 3.16). Under strong concavity of the penalty on every bounded interval, we proved that every local minimizer is finite atomic as in [20]. However, we further extend the result and prove that the number of neurons of optimal network admits an a priori bound (Theorem 3.20, Corollary 3.21). Based on that, we propose a new version of the PDAP algorithm with $n^{-1/2}$ best-iterate rate (Theorem 5.1, Corollary 5.2).
- (2) *The k -homogeneous formulation.* Extending the degree-one ReLU rescaling equivalence of Pieper and Petrosyan [20, Appendix F, Proposition 3], we show that for positively k -homogeneous activations, minimizing a separable penalty on the inner and outer parameters over radial representations induces the fractional-power penalty $|c|^{2/(k+1)}$ on the unit sphere (Lemma 4.3). The resulting problem is posed over all of $\mathcal{M}(\mathbb{S}^d)$, which has the same infimum as the finite-network problem (Proposition 4.6). We prove that a global minimizer exists, and every global minimizer is a finite network obeying an explicit a priori width bound (Theorem 4.9). We derive the atomwise optimality conditions and a positive lower bound on the weights of every total-variation local minimizer (Theorem 4.7). Every local minimizer has finite support (Corollary 4.11). An insertion algorithm is proposed based on the optimality condition (Algorithm 2).

Figure 1a compares the traditional ReLU network with an ℓ^1 penalty against representative models from the two formulations on the Van der Pol numerical reference data. At the selected checkpoints, all three plotted nonconvex models have lower error and fewer atoms than the traditional ℓ^1 model, which has 144 atoms and relative H^1 error 0.0414. These comparisons use the stated operating points, rather than an optimized baseline envelope. Figure 1b shows the corresponding accuracy–width comparison for the pendulum. The Gaussian and ReLU² attain best observed relative H^1 errors of 0.209 and 0.208, respectively, with fewer neurons than the ReLU– ℓ^1 baseline, whose error is 0.300 at 150 neurons. The softplus fit has a larger error of 0.469.

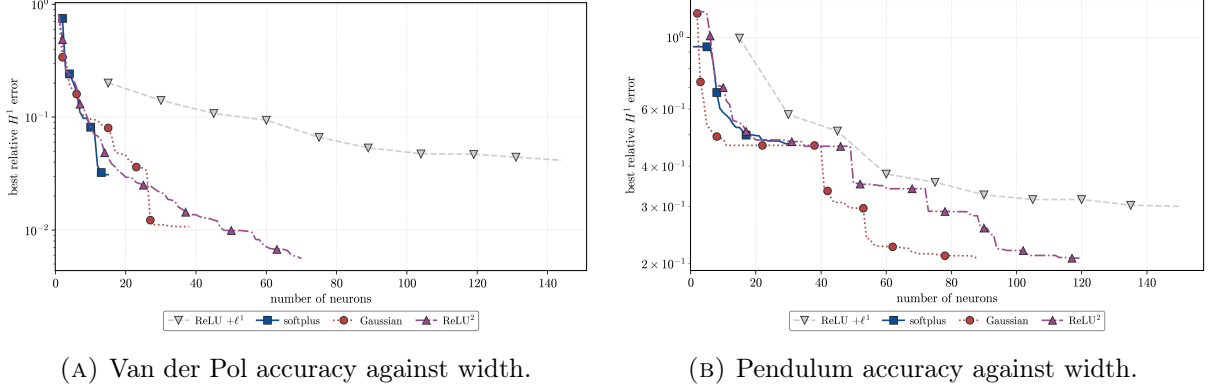


FIGURE 1. Best observed relative H^1 validation error at each support budget for (a) Van der Pol and (b) the pendulum, with the same model families and colors in both panels. The softplus and Gaussian models use Algorithm 1 with $(\alpha, \gamma, p) = (10^{-4}, 10, 2.01)$; the $\text{ReLU}-\ell^1$ baseline uses profile insertion with the linear penalty and $\alpha = 10^{-5}$; the ReLU^2 model uses ψ_2 with $\alpha = 10^{-6}$. The pendulum baseline has the lowest validation error among its five tested α values, scored at each run’s minimum-training-objective iterate; the pendulum ReLU^2 run is selected by switching-region error. Each curve is the lower envelope of validation errors over the saved iterates of that run. These are benchmark-specific operating points, not a matched- α comparison.

2. PRELIMINARIES

This section reviews the measure-theoretic facts used in the paper. The learning objectives and problem-specific assumptions are introduced in Section 3.

2.1. Finite signed Radon measures. Let Ω be a locally compact Polish space. We write $\mathcal{M}(\Omega)$ for the Banach space of finite signed Radon measures on Ω , equipped with the total-variation norm

$$\|\mu\|_{\mathcal{M}(\Omega)} := |\mu|(\Omega).$$

By the Riesz representation theorem, $\mathcal{M}(\Omega) = C_0(\Omega)^*$ under the pairing

$$\langle f, \mu \rangle := \int_{\Omega} f d\mu.$$

Here $C_0(\Omega)$ denotes the continuous functions vanishing at infinity.

For $\mu \in \mathcal{M}(\Omega)$ we define the *atom set* $\text{atom } \mu := \{\omega \in \Omega : |\mu|(\{\omega\}) > 0\}$, which is at most countable, the purely atomic part $\mu_{\text{atom}} := \sum_{\omega \in \text{atom } \mu} \mu(\{\omega\}) \delta_{\omega}$, and the nonatomic part $\mu_{\text{cont}} := \mu - \mu_{\text{atom}}$ [5]. This gives the decomposition $\mu = \mu_{\text{cont}} + \mu_{\text{atom}}$. The atom set need not be closed because atoms may accumulate at a point that carries no mass. The polar decomposition of a signed measure is written as

$$\mu = \text{sign}(\mu) |\mu|, \quad \text{sign}(\mu) := \frac{d\mu}{d|\mu|} \in \{-1, 1\} \quad |\mu| \text{-almost everywhere.}$$

2.2. Measures with a finite moment. In this subsection let $\Omega = \mathbb{R}^{d_{\Omega}}$, where $d_{\Omega} \geq 1$. For $p > 0$, set

$$w_p(\omega) := 1 + |\omega|^p$$

and define

$$(2) \quad \mathcal{M}_p(\Omega) := \left\{ \mu \in \mathcal{M}(\Omega) : \int_{\Omega} w_p d|\mu| < \infty \right\}, \quad \|\mu\|_{\mathcal{M}_p(\Omega)} := \int_{\Omega} w_p d|\mu|.$$

The associated space of continuous functions is

$$C_{w_p^{-1}}(\Omega) := \left\{ f \in C(\Omega) : \frac{f}{w_p} \in C_0(\Omega) \right\}, \quad \|f\|_{C_{w_p^{-1}}} := \sup_{\omega \in \Omega} \frac{|f(\omega)|}{w_p(\omega)}.$$

Under the dual pairing, $\mathcal{M}_p(\Omega) = C_{w_p^{-1}}(\Omega)^*$ isometrically [3, Section 5]. For an atomic measure,

$$\left\| \sum_{j=1}^N c_j \delta_{\omega_j} \right\|_{\mathcal{M}_p(\Omega)} = \sum_{\eta \in \{\omega_1, \dots, \omega_N\}} \left| \sum_{j: \omega_j = \eta} c_j \right| w_p(\eta).$$

Notation. Throughout the remainder of the paper, $D \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $\nu := \mathcal{L}^d|_D$ is the population measure, and $L^2(D)$ means $L^2(D, \nu)$. We assume that the population target V belongs to $H^1(D)$. The symbol $M \geq 1$ denotes the number of samples in an empirical loss. The inner-parameter domain is $\Omega = \mathbb{R}^{d+1}$, and $\rho \in C(\mathbb{R})$ denotes the fixed activation function of (1), with

$$\omega = (a, b), \quad [K(\omega)](x) := \rho(a \cdot x + b).$$

With this notation, the network in (1) is $\mathcal{N}_{\omega, c} = \sum_{n=1}^N c_n K(\omega_n)$. The symbol δ_ω denotes the Dirac measure at ω . We reserve p for the moment order.

3. THE LEARNING PROBLEM

3.1. The value operator and the gradient-augmented loss. The finite network $\mathcal{N}_{\omega, c}$ corresponds to the finite atomic measure $\mu_{\omega, c}$. This motivates the integral formulation below, which agrees exactly with the finite sum on atomic measures.

Assumption 3.1 (Regularity and polynomial growth of ρ). The activation satisfies $\rho \in C^1(\mathbb{R})$. There are $C_\rho > 0$, $s_0 \geq 0$, and $s_1 \geq \max\{s_0, 1\}$ such that

- (1) $|\rho(z)| \leq C_\rho(1 + |z|)^{s_0}$ for every $z \in \mathbb{R}$;
- (2) $|\rho'(z)| \leq C_\rho(1 + |z|)^{s_1-1}$ for every $z \in \mathbb{R}$.

Since D is bounded, Assumption 3.1 gives constants $C_D, C_K > 0$ such that

$$(3) \quad \|K(\omega)\|_{L^2(D)} \leq C_D(1 + |\omega|)^{s_0}.$$

Moreover, $\nabla K(a, b)(x) = a\rho'(a \cdot x + b)$, and hence

$$(4) \quad \|K(\omega)\|_{H^1(D)} \leq C_K(1 + |\omega|)^{s_1}.$$

If $\omega_n \rightarrow \omega$, their pre-activations lie in a common compact interval on D . Uniform continuity of ρ and ρ' on that interval, together with $a_n \rightarrow a$, therefore gives $K(\omega_n) \rightarrow K(\omega)$ in $H^1(D)$. Thus $K : \Omega \rightarrow H^1(D)$ is continuous and, in particular, strongly measurable by Pettis.

For $\mu \in \mathcal{M}(\Omega)$ satisfying

$$(5) \quad \int_{\Omega} \|K(\omega)\|_{H^1(D)} d|\mu|(\omega) < \infty,$$

define the $H^1(D)$ -valued Bochner integral

$$\mathcal{N}\mu := \int_{\Omega} K(\omega) d\mu(\omega).$$

By (4), condition (5) holds for $\mu \in \mathcal{M}_p(\Omega)$ when $p > s_1$. For $\psi \in C_c^\infty(D)$ and $i \in \{1, \dots, d\}$, the growth bound permits Fubini's theorem and gives

$$\begin{aligned} \int_D (\mathcal{N}\mu) \partial_i \psi dx &= \int_{\Omega} \int_D K(\omega) \partial_i \psi dx d\mu(\omega) \\ &= - \int_D \left(\int_{\Omega} \partial_i K(\omega) d\mu(\omega) \right) \psi dx. \end{aligned}$$

Consequently,

$$\nabla(\mathcal{N}\mu) = \int_{\Omega} \nabla K(\omega) d\mu(\omega) \quad \text{in } L^2(D; \mathbb{R}^d).$$

For measures satisfying only $\int_{\Omega} \|K(\omega)\|_{L^2(D)} d|\mu|(\omega) < \infty$, the same definition is valid as an $L^2(D)$ -valued Bochner integral. By (3), this includes $\mathcal{M}_p(\Omega)$ when $p > s_0$. In particular, $\mathcal{N}\mu_{\omega,c} = \mathcal{N}_{\omega,c}$. Whenever $\mathcal{N}\mu \in H^1(D)$, set $r_{\mu} := \mathcal{N}\mu - V$.

The fidelity term is the gradient-augmented loss with the distributional gradient convention. If the $L^2(D)$ -valued integral defining $\mathcal{N}\mu$ does not exist, we set $L(\mu) = +\infty$; otherwise,

$$L(\mu) := \begin{cases} \frac{1}{2} \|\mathcal{N}\mu - V\|_{L^2(D)}^2 + \frac{1}{2} \|\nabla(\mathcal{N}\mu) - \nabla V\|_{L^2(D)}^2, & \mathcal{N}\mu \in H^1(D), \\ +\infty, & \text{otherwise.} \end{cases}$$

For sampled labels $(V(x^m), g^m)_{m=1}^M$ and a finite network for which the displayed point values exist, the corresponding empirical fidelity is

$$(6) \quad l^M(\mu) := \frac{1}{2M} \sum_{m=1}^M (|\mathcal{N}\mu(x^m) - V(x^m)|^2 + |\nabla \mathcal{N}\mu(x^m) - g^m|^2).$$

3.2. The nonconvex functional. Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfy the following assumptions.

Assumption 3.2. The finite right derivative

$$L_{\phi} := \phi'(0+)$$

exists and satisfies $0 < L_{\phi} < \infty$. We use the convention $\phi'(0) := L_{\phi}$.

Assumption 3.3. The function ϕ is concave, nondecreasing, and coercive, with $\phi(0) = 0$, and belongs to $C^1((0, \infty))$.

Assumptions 3.2 and 3.3 imply that $\phi'(t) > 0$ for every $t > 0$. Indeed, if $\phi'(t_0) = 0$ at some $t_0 > 0$, concavity and monotonicity would give $\phi'(t) = 0$ for every $t \geq t_0$, making ϕ constant there and contradicting coercivity. Hence ϕ is continuous and strictly increasing from $[0, \infty)$ onto $[0, \infty)$. We denote its inverse by ϕ^{-1} .

Assumption 3.4 (Local curvature). The function ϕ belongs to $C^2((0, \infty))$ and satisfies both of the following conditions:

- (1) there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for $0 \leq z_1 \leq z_2 \leq \hat{z}_1$;
- (2) for every $A > 0$,

$$\gamma_A := - \sup_{0 < t \leq A} \phi''(t) > 0.$$

Remark 3.5. The lower curvature control in Assumption 3.4(1) is local and does not follow from global concavity. For example, $\phi(z) = z - \frac{2}{3}z^{3/2}$ near the origin has $\phi''(z) = -\frac{1}{2}z^{-1/2}$; it can be extended linearly after a small positive point so as to remain nondecreasing, concave, and coercive.

The following result follows from standard concavity estimates; see [20].

Lemma 3.6. *Let $\phi : [0, \infty) \rightarrow [0, \infty)$ be concave with $\phi(0) = 0$. Then*

- (1) $\phi(z_1 + z_2) \leq \phi(z_1) + \phi(z_2)$ for all $z_1, z_2 \geq 0$.

If in addition $L_{\phi} := \phi'(0+)$ is finite, then

- (2) $\phi(z) \leq L_{\phi}z$ for all $z \geq 0$;
- (3) $\phi(z)/z \rightarrow L_{\phi}$ as $z \downarrow 0$;

(4) for every nonnegative sequence $(t_j)_{j \geq 1}$ with $\sum_{j \geq 1} t_j < \infty$,

$$\phi\left(\sum_{j \geq 1} t_j\right) \leq \sum_{j \geq 1} \phi(t_j).$$

Proof. Concavity and $\phi(0) = 0$ make $z \mapsto \phi(z)/z$ nonincreasing on $(0, \infty)$: for $0 < z_1 \leq z_2$, writing $z_1 = \frac{z_1}{z_2}z_2 + (1 - \frac{z_1}{z_2}) \cdot 0$ and using concavity together with $\phi(0) = 0$ gives $\phi(z_1) \geq \frac{z_1}{z_2}\phi(z_2)$.

(1) Suppose without loss of generality that $0 < z_1 \leq z_2$. Then

$$\begin{aligned} \phi(z_1 + z_2) &\leq \frac{z_1 + z_2}{z_2} \phi(z_2) && \text{because } \phi(z)/z \text{ is nonincreasing} \\ &= \phi(z_2) + z_1 \frac{\phi(z_2)}{z_2} \\ &\leq \phi(z_2) + \phi(z_1) && \text{by the same monotonicity.} \end{aligned}$$

The cases with $z_1 = 0$ or $z_2 = 0$ follow from $\phi(0) = 0$.

(2) and (3) By the same monotonicity and $\phi(0) = 0$,

$$\frac{\phi(z)}{z} \leq \lim_{s \downarrow 0} \frac{\phi(s)}{s} = \lim_{s \downarrow 0} \frac{\phi(s) - \phi(0)}{s} = L_\phi \quad (z > 0),$$

and the displayed limit is precisely (3).

(4) By (2) the function ϕ is continuous at 0, and a finite concave function is continuous on $(0, \infty)$. Iterating (1) over partial sums gives

$$\phi\left(\sum_{j \leq N} t_j\right) \leq \sum_{j \leq N} \phi(t_j) \leq \sum_{j \geq 1} \phi(t_j),$$

the last step because the terms are nonnegative. Letting $N \rightarrow \infty$ and using continuity of ϕ at $\sum_{j \geq 1} t_j$ proves the claim. $\square$

Assumption 3.3 implies the standing hypotheses of Lemma 3.6, and Assumption 3.2 supplies the finite right slope needed for parts (2)–(4). Neither coercivity nor differentiability on $(0, \infty)$ is used, so the lemma also applies under the weaker hypotheses of Theorem 3.9.

Proposition 3.7 (Second-order bounds). *Suppose that $\phi(0) = 0$ and Assumption 3.2 holds.*

- (1) *If the derivative bound in Assumption 3.4(1) holds, then $\phi(z) \geq L_\phi z - \frac{\gamma_1}{2} z^2$ for $0 \leq z \leq \hat{z}_1$;*
- (2) *if $\phi \in C^2((0, \infty))$ and Assumption 3.4(2) holds, then, for every $A > 0$, $\phi(z) \leq L_\phi z - \frac{\gamma_A}{2} z^2$ for $0 \leq z \leq A$.*

Proof. For $\varepsilon > 0$, apply the fundamental theorem of calculus on $[\varepsilon, z]$ and then let $\varepsilon \downarrow 0$. The definition of the finite right derivative and $\phi(0) = 0$ give

$$\phi(z) - L_\phi z = \int_0^z [\phi'(s) - L_\phi] ds.$$

The first conclusion follows from Assumption 3.4(1); see also [20].

For the second, Assumption 3.4(2) gives $\phi''(t) \leq -\gamma_A$ on $(0, A]$. Integrating from ε to s and then letting $\varepsilon \downarrow 0$ gives $\phi'(s) - L_\phi \leq -\gamma_A s$ for $0 < s \leq A$. Integration proves the claim. $\square$

For an open set $U \subset \Omega$, let $C_c(U)$ denote the continuous functions with compact support in U . Define

$$(7) \quad \Phi_\phi(\mu) := L_\phi \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \mu} \phi(|\mu(\{\omega\})|), \quad \mu \in \mathcal{M}(\Omega).$$

We give a self-contained representation and proof of weak-* lower semicontinuity for this specialization of the nonconvex functional in Bouchitté–Buttazzo [5, Theorem 3.3 and Remark 3.5].

Lemma 3.8. *Let $U \subset \Omega = \mathbb{R}^{d+1}$ be open, let $\mu \in \mathcal{M}(U)$, and let $\delta > 0$. Suppose that*

$$|\mu(\{\omega\})| \leq \frac{\delta}{4} \quad \text{for every } \omega \in U.$$

Then, for every $\varepsilon > 0$, there are $m \geq 1$ and functions $h_1, \dots, h_m \in C_c(U)$ with pairwise disjoint supports and $|h_i| \leq 1$ such that

$$\left| \int_U h_i d\mu \right| \leq \delta \quad (i = 1, \dots, m), \quad \sum_{i=1}^m \left| \int_U h_i d\mu \right| \geq \|\mu\|_{\mathcal{M}(U)} - \varepsilon.$$

Proof. If $\varepsilon \geq \|\mu\|_{\mathcal{M}(U)}$, take $m = 1$ and $h_1 = 0$. Henceforth assume $0 < \varepsilon < \|\mu\|_{\mathcal{M}(U)}$. Let $U = P \cup N$ be a Hahn decomposition of μ . By inner regularity, there are disjoint compact sets $K_+ \subset P$ and $K_- \subset N$ such that

$$(8) \quad |\mu|(U \setminus (K_+ \cup K_-)) < \frac{\varepsilon}{4}.$$

We now construct the required finite small-mass partition, including both the atomic and nonatomic parts. For every $\omega \in K_+ \cup K_-$, continuity from above gives

$$|\mu|(B_r(\omega)) \rightarrow |\mu|(\{\omega\}) \leq \frac{\delta}{4} \quad (r \downarrow 0).$$

Choose $r_\omega > 0$ such that $|\mu|(B_{r_\omega}(\omega)) < \delta/2$. Compactness gives a finite subcover. Let $B_1, \dots, B_N$ be these balls. For each $\sigma \in \{+, -\}$, define inductively

$$E_i^\sigma := (K_\sigma \cap B_i) \setminus \bigcup_{j < i} (K_\sigma \cap B_j), \quad i = 1, \dots, N,$$

where the union is empty for $i = 1$. After discarding empty sets, the E_i^σ form a pairwise disjoint Borel partition of K_σ , with $E_i^\sigma \subset B_i$. A finite Borel partition $E_1, \dots, E_{m_0}$ for some $m_0 \geq 1$ of $K_+ \cup K_-$ is constructed such that each E_i lies in either P or N and $|\mu|(E_i) < \delta/2$.

By inner regularity, choose compact sets $C_i \subset E_i$ whose total omitted mass is less than $\varepsilon/4$, and discard the empty sets. At least one set remains because $\varepsilon < \|\mu\|_{\mathcal{M}(U)}$ and (8). Relabel the remaining sets $C_1, \dots, C_m$. They are pairwise disjoint and satisfy

$$(9) \quad \sum_{i=1}^m |\mu|(C_i) \geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4}, \quad |\mu|(C_i) \leq \frac{\delta}{2}.$$

Each C_i lies entirely in either P or N .

Because the compact sets C_i are finitely many, pairwise disjoint, and contained in the open set U , they have pairwise disjoint relatively compact open neighborhoods. Outer regularity lets us shrink these neighborhoods, while keeping each C_i inside, to obtain open sets U_i satisfying

$$(10) \quad \sum_{i=1}^m |\mu|(U_i \setminus C_i) < \min \left\{ \frac{\varepsilon}{4}, \frac{\delta}{2} \right\}.$$

For each i , the continuous function

$$\chi_i(\omega) := \frac{\text{dist}(\omega, U_i^c)}{\text{dist}(\omega, C_i) + \text{dist}(\omega, U_i^c)}$$

equals one on C_i , vanishes outside U_i , and has compact support in U . Set $h_i = \chi_i$ when $C_i \subset P$ and $h_i = -\chi_i$ when $C_i \subset N$. Their supports are pairwise disjoint and $|h_i| \leq 1$. Moreover,

$$\left| \int_U h_i d\mu \right| \leq |\mu|(U_i) \leq |\mu|(C_i) + |\mu|(U_i \setminus C_i) \leq \delta, \quad \text{by (9)–(10),}$$

and

$$\sum_{i=1}^m \left| \int_U h_i d\mu \right| \geq \sum_{i=1}^m (|\mu|(C_i) - |\mu|(U_i \setminus C_i))$$

$$\begin{aligned}
&\geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} && \text{by (9)–(10),} \\
&\geq \|\mu\|_{\mathcal{M}(U)} - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} - \frac{\varepsilon}{4} && \text{by (8),} \\
&= \|\mu\|_{\mathcal{M}(U)} - \frac{3\varepsilon}{4}.
\end{aligned}$$

This is stronger than the required lower bound. $\square$

Theorem 3.9 (Representation formula for the nonconvex functional). *Suppose that $\phi : [0, \infty) \rightarrow [0, \infty)$ is nondecreasing and concave, $\phi(0) = 0$, and*

$$L_\phi := \phi'(0+) \in (0, \infty).$$

Then, for every $\mu \in \mathcal{M}(\Omega)$,

$$(11) \quad \Phi_\phi(\mu) = \sup \left\{ \sum_{i=1}^m \phi \left(\left| \int_\Omega f_i d\mu \right| \right) : \begin{array}{l} m \geq 1, \quad f_i \in C_c(\Omega), \quad |f_i| \leq 1, \\ \text{supp}(f_i) \text{ are pairwise disjoint} \end{array} \right\}.$$

Proof. We prove the representation by first bounding every admissible finite sum from above and then constructing sums that approach $\Phi_\phi(\mu)$ from below. The hypotheses are exactly the standing hypotheses of Lemma 3.6 together with $L_\phi \in (0, \infty)$, so parts (1)–(3) of that lemma give

$$(12) \quad \phi(r+s) \leq \phi(r) + \phi(s), \quad \phi(r) \leq L_\phi r, \quad \frac{\phi(r)}{r} \longrightarrow L_\phi \quad (r \downarrow 0),$$

and part (4) gives the countable form $\phi(\sum_{j \geq 1} t_j) \leq \sum_{j \geq 1} \phi(t_j)$.

Write the atomic part of μ as

$$\mu_{\text{atom}} = \sum_{j \geq 1} c_j \delta_{\omega_j},$$

where the ω_j are distinct and $c_j \neq 0$. For an admissible family $(f_i)_{i=1}^m$, set

$$A_i := \sum_{j \geq 1} |c_j| |f_i(\omega_j)|, \quad B_i := \int_\Omega |f_i| d|\mu_{\text{cont}}|.$$

$$\begin{aligned}
\phi \left(\left| \int_\Omega f_i d\mu \right| \right) &\leq \phi(A_i + B_i) && \text{by the triangle inequality} \\
&\leq \sum_{\omega_j \in \text{supp}(f_i)} \phi(|c_j|) + L_\phi B_i && \text{by Lemma 3.6(2), (4) and } |f_i| \leq 1.
\end{aligned}$$

The disjoint supports count each atom at most once and satisfy $\sum_i |f_i| \leq 1$. Therefore

$$\begin{aligned}
\sum_{i=1}^m \phi \left(\left| \int_\Omega f_i d\mu \right| \right) &\leq \sum_{j \geq 1} \phi(|c_j|) + L_\phi \int_\Omega \sum_{i=1}^m |f_i| d|\mu_{\text{cont}}| \\
&\leq \sum_{j \geq 1} \phi(|c_j|) + L_\phi \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} = \Phi_\phi(\mu).
\end{aligned}$$

This proves the upper bound in (11).

For the reverse inequality, the claim is immediate when $\mu = 0$. Otherwise, fix $\varepsilon > 0$. Choose $\theta \in (0, L_\phi)$ such that $\theta \|\mu\|_{\mathcal{M}(\Omega)} < \varepsilon/4$. By the last limit in (12), choose $\delta > 0$ such that

$$(13) \quad \phi(r) \geq (L_\phi - \theta)r \quad (0 \leq r \leq \delta).$$

Let

$$F := \{j \geq 1 : |c_j| > \delta/8\}, \quad \mu_{\text{rem}} := \mu - \sum_{j \in F} c_j \delta_{\omega_j}.$$

The set F is finite and every point mass of μ_{rem} has magnitude at most $\delta/8$.

Because F is finite, its points have pairwise disjoint relatively compact neighborhoods. Moreover, $\mu_{\text{rem}}(\{\omega_j\}) = 0$ for $j \in F$, so continuity of the measure from above allows these neighborhoods to

be chosen with arbitrarily small total $|\mu_{\text{rem}}|$ -mass. Using the distance-function construction from Lemma 3.8, choose $g_j \in C_c(\Omega)$, $j \in F$, with pairwise disjoint supports, $|g_j| \leq 1$, and

$$g_j(\omega_j) = \text{sign}(c_j), \quad \sum_{j \in F} |\mu_{\text{rem}}|(\text{supp}(g_j)) < \tau,$$

where $\tau > 0$ can be chosen arbitrarily small. For every $j \in F$,

$$\left| \left| \int_{\Omega} g_j d\mu \right| - |c_j| \right| \leq |\mu_{\text{rem}}|(\text{supp}(g_j)).$$

By continuity of ϕ and finiteness of F , first choose τ sufficiently small and then choose the supports so that

$$(14) \quad \sum_{j \in F} \phi \left(\left| \int_{\Omega} g_j d\mu \right| \right) \geq \sum_{j \in F} \phi(|c_j|) - \frac{\varepsilon}{4}, \quad 2L_{\phi}\tau < \frac{\varepsilon}{4}.$$

Set

$$U := \Omega \setminus \bigcup_{j \in F} \text{supp}(g_j) \quad \text{and} \quad \lambda := \mu_{\text{rem}}|_U.$$

Lemma 3.8 applied on the open set U gives $h_1, \dots, h_m \in C_c(U)$ whose zero extensions belong to $C_c(\Omega)$, have pairwise disjoint supports, and satisfy

$$\left| \int_U h_i d\lambda \right| \leq \delta, \quad \sum_{i=1}^m \left| \int_U h_i d\lambda \right| \geq \|\lambda\|_{\mathcal{M}(U)} - \tau \geq \|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau.$$

Together with the g_j , the zero extensions form an admissible family in (11). Since every h_i is supported in U and $\mu|_U = \lambda$, its integrals against μ and λ agree. Therefore (13) and (14) yield

$$\begin{aligned} & \sum_{j \in F} \phi \left(\left| \int_{\Omega} g_j d\mu \right| \right) + \sum_{i=1}^m \phi \left(\left| \int_{\Omega} h_i d\mu \right| \right) \\ & \geq \sum_{j \in F} \phi(|c_j|) - \frac{\varepsilon}{4} + (L_{\phi} - \theta)(\|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau) \\ & \geq \Phi_{\phi}(\mu) - \frac{\varepsilon}{4} - \theta\|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} - 2L_{\phi}\tau \quad \text{by } \Phi_{\phi}(\mu_{\text{rem}}) \leq L_{\phi}\|\mu_{\text{rem}}\|_{\mathcal{M}(\Omega)} \\ & > \Phi_{\phi}(\mu) - \varepsilon. \end{aligned}$$

Taking the supremum and then letting $\varepsilon \downarrow 0$ proves the reverse inequality. $\square$

Corollary 3.10. *Under the assumptions of Theorem 3.9, the functional Φ_{ϕ} is weak-* lower semicontinuous on $\mathcal{M}(\Omega) = C_0(\Omega)^*$. If ϕ is also coercive, then every sublevel of Φ_{ϕ} is bounded in total variation.*

Proof. For each fixed admissible family in (11), weak-* convergence gives convergence of all its integrals, and continuity of ϕ gives convergence of the corresponding finite sum. The representation is a supremum of weak-* continuous functions and is therefore weak-* lower semicontinuous.

For the sublevel bound, apply repeated subadditivity to the first N atoms and let $N \rightarrow \infty$. The atom magnitudes are summable, so continuity of ϕ gives

$$\phi(\|\mu_{\text{atom}}\|_{\mathcal{M}(\Omega)}) \leq \sum_{\omega \in \text{atom } \mu} \phi(|\mu(\{\omega\})|) \leq \Phi_{\phi}(\mu), \quad L_{\phi}\|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} \leq \Phi_{\phi}(\mu).$$

If $\Phi_{\phi}(\mu) \leq C$, coercivity bounds $\|\mu_{\text{atom}}\|_{\mathcal{M}(\Omega)}$ in terms of C , while $\|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} \leq C/L_{\phi}$. The atomic part is supported on the countable atom set, while the continuous part assigns that set zero variation. The two parts are therefore mutually singular, so their total-variation norms add to $\|\mu\|_{\mathcal{M}(\Omega)}$. Hence the sublevel is bounded in total variation. $\square$

3.3. The normalized problem. For $\alpha > 0$, define

$$J_0(\mu) := L(\mu) + \alpha \Phi_\phi(\mu)$$

for the unnormalized problem on the parameter domain.

Escape on the unbounded parameter domain. For positively homogeneous profiles the radial scale of ω is absorbed by the outer coefficient, and the parameter can be restricted to the unit sphere $\mathbb{S}^d$ (Section 4). But in general the domain is $\Omega = \mathbb{R}^{d+1}$, which is not compact. The regularizer Φ_ϕ depends on the sizes of the coefficients but not on the locations of the atoms. Along a sequence with $|\omega| \rightarrow \infty$, the fidelity need not be lower semicontinuous.

Example 3.11 (Failure of weak-* lower semicontinuity for tanh). Let $\rho = \tanh$, $V \equiv 1$, fix $a \in \mathbb{R}^d$, and set $\mu_n = \delta_{(a, b_n)}$ with $b_n \rightarrow \infty$. Then $\mu_n \xrightarrow{*} 0$, yet

$$\mathcal{N}\mu_n = \tanh(a \cdot x + b_n) \rightarrow 1$$

strongly in $H^1(D)$. Hence $\liminf_{n \rightarrow \infty} L(\mu_n) = 0 < L(0)$, so the fidelity term is not weak-* lower semicontinuous. Mass escaping to infinity carries a non-vanishing value contribution unless $K(\omega)$ tends to zero as $|\omega| \rightarrow \infty$.

Note that $\Phi_\phi(\mu_n) = \phi(1)$ is constant along this sequence, so the unnormalized coefficient penalty does not prevent this escape. The defect is that the unnormalized kernel does not vanish at parameter infinity. The normalized atom introduced in Theorem 3.13 does vanish under the corresponding polynomial growth gap.

Example 3.12 (Nonattainment for softplus). For the profile $\rho(z) = \log(1 + e^z)$ (softplus), let $b_n \rightarrow \infty$. The measures $\mu_n = \rho(b_n)^{-1} \delta_{(0, b_n)}$ fit $V \equiv 1$ exactly for every n at penalty $\alpha \phi(1/\rho(b_n)) \rightarrow 0$, so $\inf J_0 = 0$ is not attained. Indeed, if $J_0(\mu) = 0$, then $\Phi_\phi(\mu) = 0$. Strict positivity of ϕ on $(0, \infty)$ and $L_\phi > 0$ force both the atomic and continuous parts of μ to vanish. Thus $\mu = 0$, but $L(0) = \frac{1}{2} \|1\|_{H^1(D)}^2 > 0$, a contradiction.

We need a penalty that is aware of the location of the atom to prevent the escape. The space of measure of finite moment is the right space to work with. For $p > 0$, let w_p and $\mathcal{M}_p(\Omega)$ be as in (2). For $\mu \in \mathcal{M}_p(\Omega)$, we define the normalized measure and normalized atom by

$$d\mu_p := w_p d\mu, \quad K_p(\omega) := \frac{K(\omega)}{w_p(\omega)}.$$

Theorem 3.13. *Multiplication by w_p is a linear isometric bijection*

$$\mathcal{M}_p(\Omega) \longrightarrow \mathcal{M}(\Omega), \quad \mu \longmapsto \mu_p,$$

with inverse $d\mu = w_p^{-1} d\mu_p$, and

$$(15) \quad \|\mu\|_{\mathcal{M}_p(\Omega)} = \|\mu_p\|_{\mathcal{M}(\Omega)}.$$

Moreover, μ_n converges weak-* to μ in $C_{w_p^{-1}}(\Omega)^*$ if and only if $(\mu_n)_p \xrightarrow{*} \mu_p$ in $C_0(\Omega)^*$. The multiplication preserves the atom set and the polar sign, and

$$(16) \quad (\mu_p)_{\text{atom}} = w_p \mu_{\text{atom}}, \quad (\mu_p)_{\text{cont}} = w_p \mu_{\text{cont}}.$$

If part (1) of Assumption 3.1 holds and $p > s_0$, then $K_p \in C_0(\Omega; L^2(D))$ and

$$(17) \quad \mathcal{N}\mu = \int_{\Omega} K d\mu = \int_{\Omega} K_p d\mu_p.$$

Finally,

$$(18) \quad \Phi_\phi(\mu_p) = L_\phi \int_{\Omega} w_p d|\mu_{\text{cont}}| + \sum_{\omega \in \text{atom } \mu} \phi(w_p(\omega) |\mu(\{\omega\})|).$$

Proof. The isometric identification with $\mathcal{M}(\Omega)$, including its inverse, the norm identity, and the weak-* equivalence, is the polynomial-weight specialization of [3, Section 5.1]; in particular, this gives (15). Since multiplication by the strictly positive function w_p preserves null sets, the atom set, the polar sign, and nonatomicity, (16) follows immediately.

It remains to treat the network map. The continuity of K_p follows from that of K and w_p . For $r = |\omega| \geq 1$, the growth estimate (3) gives

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\leq C_D \frac{(1+r)^{s_0}}{1+r^p} \\ &\leq 2^{s_0} C_D r^{s_0-p} \longrightarrow 0, \quad r \rightarrow \infty, \end{aligned}$$

because $p > s_0$. Thus $K_p \in C_0(\Omega; L^2(D))$. The change-of-measure identity for the Bochner integral now gives (17). Finally, (16) and $|w_p \mu_{\text{cont}}| = w_p |\mu_{\text{cont}}|$ give (18) directly from (7). $\square$

For $\alpha > 0$, define the adopted objective

$$J(\mu) := L(\mu) + \alpha \Phi_\phi(\mu_p), \quad \mu \in \mathcal{M}_p(\Omega).$$

For a finite atomic measure $\mu_{\omega,c} = \sum_{n=1}^N c_n \delta_{\omega_n}$ with distinct support points, this becomes

$$J(\mu_{\omega,c}) = L(\mu_{\omega,c}) + \alpha \sum_{n=1}^N \phi(w_p(\omega_n)|c_n|).$$

Throughout, finite-network representations are understood after repeated locations are merged and zero coefficients are discarded; their width is therefore $\# \text{ atom } \mu$.

Definition 3.14 (Local minimizer). A measure $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is a local minimizer of J if $J(\bar{\mu}) < \infty$ and there is $\varepsilon > 0$ such that

$$J(\bar{\mu}) \leq J(\mu) \quad \text{for all} \quad \mu \in \mathcal{M}_p(\Omega), \quad \|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} < \varepsilon.$$

By Theorem 3.13, this is exactly total-variation local minimality in the normalized coordinate μ_p .

3.4. Existence of a global minimizer. We apply the direct method in the normalized coordinate. Weak-* compactness controls μ_p , Corollary 3.10 supplies the penalty lower semicontinuity and sublevel bound, and Lemma 3.15 gives the strong convergence needed for the value part of the fidelity.

Lemma 3.15. Assume (3), let $p > s_0$, and suppose that

$$(\mu_n)_p \xrightarrow{*} \mu_p \quad \text{in } \mathcal{M}(\Omega).$$

Then $\mathcal{N}\mu_n \rightarrow \mathcal{N}\mu$ strongly in $L^2(D)$.

Proof. We use a compactly supported cutoff to pass to the weak-* limit locally and then control the tails uniformly. By Theorem 3.13,

$$\mathcal{N}\mu_n = \int_{\Omega} K_p d(\mu_n)_p, \quad \mathcal{N}\mu = \int_{\Omega} K_p d\mu_p.$$

Set

$$C := \sup_n \|(\mu_n)_p\|_{\mathcal{M}(\Omega)} < \infty,$$

where finiteness follows from weak-* convergence and the uniform boundedness principle. Weak-* lower semicontinuity of total variation gives $\|\mu_p\|_{\mathcal{M}(\Omega)} \leq C$.

For $R \geq 1$, choose $\chi_R \in C_c(\Omega)$ with $0 \leq \chi_R \leq 1$, $\chi_R = 1$ on B_R , and $\text{supp } \chi_R \subset B_{R+1}$. For fixed R and $x \in D$, the function

$$\omega \longmapsto \chi_R(\omega)[K_p(\omega)](x)$$

belongs to $C_c(\Omega)$. Hence weak-* convergence gives

$$\int_{\Omega} \chi_R(\omega)[K_p(\omega)](x) d(\mu_n)_p(\omega) \longrightarrow \int_{\Omega} \chi_R(\omega)[K_p(\omega)](x) d\mu_p(\omega)$$

for every $x \in D$.

Since D is bounded and $\text{supp } \chi_R$ is compact,

$$M_R := \sup_{\omega \in \text{supp } \chi_R, x \in D} |[K_p(\omega)](x)| < \infty.$$

The absolute values of both truncated integrals are bounded by CM_R . The integrand $(\omega, x) \mapsto \chi_R(\omega)[K_p(\omega)](x)$ is jointly Borel measurable and bounded by M_R ; Fubini's theorem therefore identifies the pointwise integrals above, almost everywhere in x , with representatives of the $L^2(D)$ -valued Bochner integrals below. The constant function $2CM_R$ belongs to $L^2(D)$ because D is bounded. Dominated convergence therefore yields

$$\int_{\Omega} \chi_R K_p d(\mu_n)_p \longrightarrow \int_{\Omega} \chi_R K_p d\mu_p \quad \text{strongly in } L^2(D).$$

It remains to remove the cutoff. If $r := |\omega| > R \geq 1$, then

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\leq C_D \frac{(1+r)^{s_0}}{1+r^p} && \text{by (3)} \\ &\leq 2^{s_0} C_D r^{s_0-p} \leq 2^{s_0} C_D R^{s_0-p}, && \text{since } p > s_0 \text{ and } r > R. \end{aligned}$$

Consequently,

$$\begin{aligned} \left\| \int_{\Omega} (1 - \chi_R) K_p d(\mu_n)_p \right\|_{L^2(D)} &\leq 2^{s_0} C_D C R^{s_0-p}, \\ \left\| \int_{\Omega} (1 - \chi_R) K_p d\mu_p \right\|_{L^2(D)} &\leq 2^{s_0} C_D C R^{s_0-p}. \end{aligned}$$

For fixed R , let $n \rightarrow \infty$ in the truncated integrals. Then let $R \rightarrow \infty$; the two tail bounds vanish because $s_0 - p < 0$. This proves the claim. $\square$

Theorem 3.16 (Existence on the unbounded parameter domain). *Let $V \in H^1(D)$, let ϕ satisfy Assumptions 3.2 and 3.3, and let $\rho \in C(\mathbb{R})$ satisfy Assumption 3.1(1). If $p > s_0$ and $\alpha > 0$, then*

$$\inf_{\mu \in \mathcal{M}_p(\Omega)} J(\mu)$$

admits at least one global minimizer.

Proof. We use the direct method in the normalized measure coordinate. Only the stated continuity and value-growth bound for ρ are needed. Since $V \in H^1(D)$,

$$J(0) = \frac{1}{2} \|V\|_{H^1(D)}^2 =: C < \infty.$$

Choose a minimizing sequence $(\mu_n) \subset \mathcal{M}_p(\Omega)$ with $J(\mu_n) \leq C + 1$. The fidelity and penalty are nonnegative, so

$$\Phi_{\phi}((\mu_n)_p) \leq \frac{C+1}{\alpha}.$$

The total-variation sublevel bound in Corollary 3.10 shows that $((\mu_n)_p)$ is bounded in $\mathcal{M}(\Omega)$. Weak-* compactness and separability of $C_0(\Omega)$ give, after passing to a subsequence,

$$(\mu_n)_p \xrightarrow{*} \bar{\mu}_p.$$

Theorem 3.13 defines a unique $\bar{\mu} \in \mathcal{M}_p(\Omega)$ with normalized coordinate $\bar{\mu}_p$. Lemma 3.15 gives

$$(19) \quad \mathcal{N}_{\mu_n} \longrightarrow \mathcal{N}_{\bar{\mu}} \quad \text{strongly in } L^2(D).$$

The fidelity bound gives

$$\sup_n \|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)} \leq \|\nabla V\|_{L^2(D)} + \sqrt{2(C+1)}.$$

After another subsequence, $\nabla(\mathcal{N}\mu_n) \rightharpoonup g$ weakly in $L^2(D)^d$. For every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi \, dx = \lim_{n \rightarrow \infty} \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi \, dx = - \lim_{n \rightarrow \infty} \int_D \mathcal{N}\mu_n \operatorname{div} \psi \, dx = - \int_D \mathcal{N}\bar{\mu} \operatorname{div} \psi \, dx,$$

the last step by (19). Hence $g = \nabla(\mathcal{N}\bar{\mu})$ distributionally and $\mathcal{N}\bar{\mu} \in H^1(D)$.

The value part of the fidelity is continuous by (19), and the gradient part is weakly lower semicontinuous. Hence

$$L(\bar{\mu}) \leq \liminf_{n \rightarrow \infty} L(\mu_n).$$

Corollary 3.10 gives

$$\Phi_\phi(\bar{\mu}_p) \leq \liminf_{n \rightarrow \infty} \Phi_\phi((\mu_n)_p).$$

Combining the two inequalities proves $J(\bar{\mu}) \leq \liminf_n J(\mu_n) = \inf J$. $\square$

3.5. Optimality conditions and finite support. Let $\bar{\mu}$ be a local minimizer. For a measure μ and an admissible direction μ' , the Gâteaux derivative is

$$DL(\mu)[\mu'] := \lim_{t \rightarrow 0} \frac{L(\mu + t\mu') - L(\mu)}{t} = \langle r_\mu, \mathcal{N}\mu' \rangle_{H^1(D)}.$$

The corresponding derivative profile is

$$P_\mu(\omega) := DL(\mu)[\delta_\omega] = \langle r_\mu, K(\omega) \rangle_{H^1(D)}.$$

Define its normalized form and the uniform normalized feature bound by

$$P_{p,\mu}(\omega) := \frac{P_\mu(\omega)}{w_p(\omega)} = \langle r_\mu, K_p(\omega) \rangle_{H^1(D)}, \quad B_p := \sup_{\omega \in \Omega} \|K_p(\omega)\|_{H^1(D)}.$$

At the local minimizer write $\bar{P} := P_{\bar{\mu}}$ and $\bar{P}_p := P_{p,\bar{\mu}}$. Thus $\bar{P}$ is the function representing the Gâteaux derivative. For differentiable ρ ,

$$\bar{P}(a, b) = \int_D r_{\bar{\mu}}(x) \rho(a \cdot x + b) \, dx + \int_D \nabla r_{\bar{\mu}}(x) \cdot a \rho'(a \cdot x + b) \, dx.$$

The second term carries the factor a , so on the unbounded domain $\bar{P}$ is defined pointwise but need not be bounded.

By Cauchy–Schwarz, (4) gives

$$|\bar{P}(\omega)| \leq \|r_{\bar{\mu}}\|_{H^1(D)} \|K(\omega)\|_{H^1(D)} \leq C_P(1 + |\omega|)^{s_1}, \quad C_P := C_K \|r_{\bar{\mu}}\|_{H^1(D)}.$$

The constant C_P depends on the fixed minimizer through its residual, but not on ω . If $p > s_1$, then

$$K_p \in C_0(\Omega; H^1(D)), \quad \bar{P}_p \in C_0(\Omega), \quad \bar{P} \in C_{w_p^{-1}}(\Omega).$$

Consequently, for every $\mu' \in \mathcal{M}_p(\Omega)$,

$$DL(\bar{\mu})[\mu'] = \int_\Omega \bar{P} \, d\mu' = \int_\Omega \bar{P}_p \, d\mu'_p.$$

The pointwise definition of $\bar{P}$ as a Gâteaux derivative does not require $\bar{P} \in C_0(\Omega)$.

Theorem 3.17 (The optimality condition). *Let $\phi : [0, \infty) \rightarrow [0, \infty)$ be concave and nondecreasing, with $\phi(0) = 0$, $\phi \in C^1((0, \infty))$, and $0 < L_\phi := \phi'(0+) < \infty$. Let $\bar{\mu} \in \mathcal{M}_p(\Omega)$ be a local minimizer of J with $J(\bar{\mu}) < \infty$, and suppose that Assumption 3.1 holds and $p > s_1$. Then*

$$(20) \quad |\bar{P}_p(\omega)| \leq \alpha L_\phi \quad \text{for every } \omega \in \Omega,$$

$$(21) \quad \bar{P}_p(\omega) = -\alpha \phi'(w_p(\omega) |\bar{\mu}(\{\omega\})|) \operatorname{sign}(\bar{\mu}(\{\omega\})) \quad \text{for } \omega \in \operatorname{atom} \bar{\mu}.$$

Moreover,

$$(22) \quad \bar{P}_p = -\alpha L_\phi \operatorname{sign}(\bar{\mu}_{\text{cont}}) \quad |(\bar{\mu}_p)_{\text{cont}}| \text{-almost everywhere.}$$

Proof. We use Theorem 3.13 to work with $\bar{\mu}_p$ in the total-variation norm. The normalized fidelity derivative in the direction $\xi \in \mathcal{M}(\Omega)$ is

$$\left\langle r_{\bar{\mu}}, \int_{\Omega} K_p d\xi \right\rangle_{H^1(D)} = \int_{\Omega} \bar{P}_p d\xi.$$

Fix $\omega \in \operatorname{atom} \bar{\mu}$ and put $z := \bar{\mu}_p(\{\omega\}) = w_p(\omega)\bar{\mu}(\{\omega\}) \neq 0$. For $s \in \{-1, 1\}$ and sufficiently small $t > 0$, the sign of $z + ts$ equals the sign of z . Dividing the local-minimality inequality by t and letting $t \downarrow 0$ gives

$$0 \leq s [\bar{P}_p(\omega) + \alpha \phi'(|z|) \operatorname{sign}(z)].$$

Using both choices of s makes the bracket vanish, and hence

$$\bar{P}_p(\omega) = -\alpha \phi'(|z|) \operatorname{sign}(z).$$

Because $w_p > 0$, $\operatorname{sign}(z) = \operatorname{sign}(\bar{\mu}(\{\omega\}))$, so this is (21). Concavity and monotonicity give $0 \leq \phi'(|z|) \leq L_\phi$, and hence $|\bar{P}_p(\omega)| \leq \alpha L_\phi$ at every atom.

If $\omega \notin \operatorname{atom} \bar{\mu}$, insertion of $ts\delta_\omega$ into the normalized measure changes the penalty by $\phi(t)$. Dividing the local minimality inequality by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \leq s \bar{P}_p(\omega) + \alpha L_\phi.$$

The two choices of s prove (20) away from the atoms, and the preceding atom estimate completes its proof.

It remains to identify the continuous part. Set $\xi := (\bar{\mu}_p)_{\text{cont}}$. Since $K_p \in C_0(\Omega; H^1(D))$, the directions $\pm \xi$ are admissible. For $s \in \{-1, 1\}$ and $0 < t < 1$, the perturbation $\bar{\mu}_p + ts\xi$ multiplies the continuous part by $1 + ts > 0$. Its one-sided directional derivative therefore gives

$$0 \leq s \left[\int_{\Omega} \bar{P}_p d\xi + \alpha L_\phi \|\xi\|_{\mathcal{M}(\Omega)} \right].$$

Using both signs shows that the bracket vanishes. On the other hand, (20) makes

$$\bar{P}_p \operatorname{sign}(\xi) + \alpha L_\phi \geq 0$$

$|\xi|$ -almost everywhere. Its integral is zero, so the integrand vanishes $|\xi|$ -almost everywhere. The sign preservation in Theorem 3.13 gives (22). $\square$

Remark 3.18. Because $\bar{P}_p \in C_0(\Omega)$ and $\alpha L_\phi > 0$, every closed positive superlevel set of $|\bar{P}_p|$ is compact. Thus any violation of (20) is confined to a compact set, even when the unnormalized function $\bar{P}$ does not decay at parameter infinity. Lemma 3.19 uses a lower superlevel to localize the entire support.

Lemma 3.19. *Let ϕ satisfy Assumptions 3.2 and 3.3, let $p > s_1$, and suppose that Assumption 3.1 holds. Suppose that $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is an $\mathcal{M}_p(\Omega)$ -local minimizer of J such that $0 < J(\bar{\mu}) < \infty$. Define*

$$T(\bar{\mu}) := \phi^{-1} \left(\frac{J(\bar{\mu})}{\alpha} \right)$$

and

$$\mathcal{K}(\bar{\mu}) := \{\omega \in \Omega : |\bar{P}_p(\omega)| \geq \alpha \phi'(T(\bar{\mu}))\}.$$

Then $\mathcal{K}(\bar{\mu})$ is compact and

$$(23) \quad \operatorname{supp} \bar{\mu}_{\text{atom}} \cup \operatorname{supp} \bar{\mu}_{\text{cont}} \subset \mathcal{K}(\bar{\mu}).$$

Moreover, every atom $\omega \in \operatorname{atom} \bar{\mu}$ satisfies

$$w_p(\omega) |\bar{\mu}(\{\omega\})| \leq T(\bar{\mu}).$$

Proof. We use Theorem 3.17 and the fact that $\bar{P}_p$ vanishes at infinity to prove the localization. Concavity and monotonicity make ϕ' positive and nonincreasing, as observed after Assumption 3.3. Since $\bar{P}_p \in C_0(\Omega)$, its closed superlevel set at the strictly positive level $\alpha\phi'(T(\bar{\mu}))$ is compact. This proves the first assertion.

Let $\omega \in \text{atom } \bar{\mu}$ and put

$$t_\omega := |\bar{\mu}_p(\{\omega\})| = w_p(\omega)|\bar{\mu}(\{\omega\})|.$$

The contribution of this atom to the objective gives

$$\begin{aligned} \alpha\phi(t_\omega) &\leq \alpha\Phi_\phi(\bar{\mu}_p) && \text{by nonnegativity of the other terms} \\ &\leq J(\bar{\mu}) = \alpha\phi(T(\bar{\mu})). \end{aligned}$$

Strict monotonicity of ϕ gives $t_\omega \leq T(\bar{\mu})$. The atom condition in Theorem 3.17 and monotonicity of ϕ' now give

$$|\bar{P}_p(\omega)| = \alpha\phi'(t_\omega) \geq \alpha\phi'(T(\bar{\mu})).$$

Thus every atom lies in $\mathcal{K}(\bar{\mu})$.

On the continuous part, Theorem 3.17 gives

$$|\bar{P}_p(\omega)| = \alpha L_\phi \geq \alpha\phi'(T(\bar{\mu})) \quad |(\bar{\mu}_p)_{\text{cont}}| \text{-almost everywhere.}$$

Hence $(\bar{\mu}_p)_{\text{cont}}$ is concentrated on the closed set $\mathcal{K}(\bar{\mu})$. Theorem 3.13 preserves the atomic and continuous decompositions. Since w_p is strictly positive, each component of $\bar{\mu}_p = w_p\bar{\mu}$ has the same null sets, and hence the same support, as the corresponding component of $\bar{\mu}$. Taking closures for the atomic part and supports for the continuous part proves (23). $\square$

Theorem 3.20 (Finite support and separation of local minimizers). *Let ϕ satisfy Assumptions 3.2 and 3.3, suppose that $\phi \in C^2((0, \infty))$ satisfies Assumption 3.4(2), let $p > s_1$, and suppose that Assumption 3.1 holds. Then every $\mathcal{M}_p(\Omega)$ -local minimizer $\bar{\mu} \in \mathcal{M}_p(\Omega)$ of J with $J(\bar{\mu}) < \infty$ is finite atomic. If $J(\bar{\mu}) > 0$, set*

$$T := T(\bar{\mu}), \quad \mathcal{K} := \mathcal{K}(\bar{\mu}).$$

Then $\bar{\mu}_{\text{cont}} = 0$, every atom belongs to the compact set $\mathcal{K}$, and any two distinct atoms ω_i and ω_j satisfy

$$(24) \quad \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 \geq 2\alpha\gamma T.$$

Consequently,

$$(25) \quad \#\text{atom}(\bar{\mu}) \leq N_{\text{cov}}\left(K_p(\mathcal{K}), \frac{1}{2}\sqrt{\alpha\gamma T}\right) < \infty,$$

where the covering number $N_{\text{cov}}(A, r)$ is the least number of balls of radius r needed to cover A .

Proof. We use the isometry of Theorem 3.13, the first-order identities of Theorem 3.17, and local coefficient variations to prove the assertions. Concavity, coercivity, and $L_\phi > 0$ imply that $\phi(t) > 0$ for $t > 0$.

The zero objective case is immediate:

$$J(\bar{\mu}) = 0 \implies \Phi_\phi(\bar{\mu}_p) = 0 \implies \bar{\mu}_p = 0 \implies \bar{\mu} = 0,$$

where the middle implication follows from $L_\phi > 0$ and $\phi(t) > 0$ for $t > 0$. We may therefore suppose $J(\bar{\mu}) > 0$. Lemma 3.19 gives the compact set $\mathcal{K}$ and already localizes the atomic and continuous supports. Every normalized atom magnitude

$$t_i := |\bar{\mu}_p(\{\omega_i\})|$$

satisfies $t_i \leq T$ by Lemma 3.19. The curvature assumption gives $\phi''(t) \leq -\gamma_T$ for $0 < t \leq T$. Integrating this inequality from the origin, using $\phi'(0+) = L_\phi$, and then integrating once more gives

$$(26) \quad L_\phi t - \phi(t) \geq \frac{\gamma_T}{2} t^2 \quad (0 \leq t \leq T).$$

We first separate distinct atoms. For two such atoms, Theorem 3.13 defines a unique measure $\mu_s \in \mathcal{M}_p(\Omega)$ by

$$(\mu_s)_p := \bar{\mu}_p + s\delta_{\omega_i} - s\delta_{\omega_j}.$$

For sufficiently small $|s|$, the signs of both atom coefficients remain fixed, and

$$\begin{aligned} \|\mu_s - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} &= 2|s|, & \text{by Theorem 3.13,} \\ \mathcal{N}\mu_s &= \mathcal{N}\bar{\mu} + s[K_p(\omega_i) - K_p(\omega_j)], & \text{by the normalized representation of } \mathcal{N}. \end{aligned}$$

Thus $s \mapsto J(\mu_s)$ is twice differentiable near zero and has a local minimum there. Its second derivative satisfies

$$\begin{aligned} 0 &\leq \left. \frac{d^2}{ds^2} J(\mu_s) \right|_{s=0} \\ &= \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 + \alpha\phi''(t_i) + \alpha\phi''(t_j) & \text{by the quadratic fidelity} \\ &\leq \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 - 2\alpha\gamma_T, & \text{since } t_i, t_j \leq T. \end{aligned}$$

This proves (24).

We next rule out the continuous part by concentrating a local piece of one sign. Suppose $(\bar{\mu}_p)_{\text{cont}} \neq 0$ and write

$$d(\bar{\mu}_p)_{\text{cont}} = \sigma d\lambda, \quad \lambda := |(\bar{\mu}_p)_{\text{cont}}|,$$

where σ takes the values -1 and 1 λ -almost everywhere. Choose $\sigma_0 \in \{-1, 1\}$ such that the restriction of λ to $\{\sigma = \sigma_0\}$ is nonzero, and denote that restriction again by λ . Theorem 3.17 gives $\bar{P}_p = -\alpha L_\phi \sigma_0$ on a set of full λ -measure. The support of a Radon measure has full measure, while the countable set atom $\bar{\mu}$ has zero λ -measure because λ is nonatomic. The intersection of these three full λ -measure sets is nonempty, so choose

$$\bar{\omega} \in \text{supp } \lambda \setminus \text{atom } \bar{\mu} \quad \text{such that} \quad \bar{P}_p(\bar{\omega}) = -\alpha L_\phi \sigma_0.$$

For $\delta > 0$, put

$$r_\delta := \lambda(B_\delta(\bar{\omega})), \quad \varepsilon_\delta := \sup_{\omega \in B_\delta(\bar{\omega})} \|K_p(\omega) - K_p(\bar{\omega})\|_{H^1(D)}.$$

Because $\bar{\omega} \in \text{supp } \lambda$, one has $r_\delta > 0$. Moreover, $r_\delta \rightarrow \lambda(\{\bar{\omega}\}) = 0$ by continuity from above and the nonatomicity of λ , while $\varepsilon_\delta \rightarrow 0$ by continuity of K_p at $\bar{\omega}$. Define μ_δ through its normalized coordinate by

$$(\mu_\delta)_p := \bar{\mu}_p - \sigma_0 \lambda|_{B_\delta(\bar{\omega})} + \sigma_0 r_\delta \delta_{\bar{\omega}}.$$

The target point is not an atom, and Theorem 3.13 gives

$$\|\mu_\delta - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} = 2r_\delta \rightarrow 0.$$

If $e_\delta := \mathcal{N}(\mu_\delta - \bar{\mu})$, then

$$\|e_\delta\|_{H^1(D)} \leq \varepsilon_\delta r_\delta, \quad \text{by the Bochner-integral triangle inequality,}$$

$$\langle r_{\bar{\mu}}, e_\delta \rangle_{H^1(D)} = \sigma_0 r_\delta \bar{P}_p(\bar{\omega}) - \sigma_0 \int_{B_\delta(\bar{\omega})} \bar{P}_p d\lambda = 0, \quad \text{by Theorem 3.17.}$$

The exact quadratic expansion of the fidelity and the definition of Φ_ϕ therefore give, for all sufficiently small δ such that $r_\delta \leq T$,

$$J(\mu_\delta) - J(\bar{\mu}) = \frac{1}{2} \|e_\delta\|_{H^1(D)}^2 + \alpha[\phi(r_\delta) - L_\phi r_\delta]$$

$$\begin{aligned}
&\leq \frac{1}{2}(\varepsilon_\delta^2 - \alpha\gamma_T)r_\delta^2 && \text{by (26)} \\
&< 0, && \text{for sufficiently small } \delta.
\end{aligned}$$

This contradicts local minimality. Hence $\bar{\mu}_{\text{cont}} = 0$.

It remains to count the atoms. The set $K_p(\mathcal{K})$ is compact because $\mathcal{K}$ is compact and K_p is continuous. Cover it by balls of radius $\frac{1}{2}\sqrt{\alpha\gamma_T}$. Each such ball has diameter $\sqrt{\alpha\gamma_T}$, which is strictly smaller than the separation $\sqrt{2\alpha\gamma_T}$ in (24). Hence each ball contains the image of at most one atom. A compact set has a finite cover at every positive radius, and (25) follows. $\square$

Corollary 3.21. *Assume the hypotheses of Theorems 3.16 and 3.20. Then J admits a global minimizer, every global minimizer is finite atomic, and*

$$(27) \quad \min_{\mu \in \mathcal{M}_p(\Omega)} J(\mu) = \min_{\substack{N \geq 1, \omega_1, \dots, \omega_N \in \Omega \\ c_1, \dots, c_N \in \mathbb{R}}} \left\{ \frac{1}{2} \left\| \sum_{n=1}^N c_n K(\omega_n) - V \right\|_{H^1(D)}^2 + \alpha \sum_{n=1}^N \phi(w_p(\omega_n)|c_n|) \right\}.$$

The minimum on the right is attained. If μ_* is any global minimizer with $J(\mu_*) > 0$, then it satisfies the data-dependent bound

$$(28) \quad \# \text{atom}(\mu_*) \leq N_{\text{cov}} \left(K_p(\mathcal{K}(\mu_*)), \frac{1}{2} \sqrt{\alpha\gamma_{T(\mu_*)}} \right).$$

If $V \neq 0$, define

$$T_0 := \phi^{-1} \left(\frac{\|V\|_{H^1(D)}^2}{2\alpha} \right)$$

and

$$(29) \quad \mathcal{K}_0 := \{ \omega \in \Omega : \|V\|_{H^1(D)} \|K_p(\omega)\|_{H^1(D)} \geq \alpha\phi'(T_0) \}.$$

Then $\mathcal{K}_0$ is compact, every atom of every global minimizer lies in $\mathcal{K}_0$, and

$$(30) \quad \# \text{atom}(\mu_*) \leq N_{\text{cov}} \left(K_p(\mathcal{K}_0), \frac{1}{2} \sqrt{\alpha\gamma_{T_0}} \right) < \infty.$$

If $V = 0$, then the zero measure is the unique global minimizer.

Proof. We combine the direct method in Theorem 3.16 with the local result in Theorem 3.20. Theorem 3.16 gives a global minimizer. Every global minimizer is $\mathcal{M}_p(\Omega)$ -local, so Theorem 3.20 makes it finite atomic; when its objective is positive, Theorem 3.20 also gives (28). Its distinct atom representation has objective equal to the expression on the right of (27), so the right-hand infimum is no larger than the measure minimum.

Conversely, consider an arbitrary finite network, possibly with repeated locations. For each distinct location η_j , merge its coefficients into $d_j := \sum_{n: \omega_n = \eta_j} c_n$. The represented function, and hence the fidelity, is unchanged. Monotonicity and repeated subadditivity of ϕ give

$$\phi(w_p(\eta_j)|d_j|) \leq \phi \left(w_p(\eta_j) \sum_{n: \omega_n = \eta_j} |c_n| \right) \leq \sum_{n: \omega_n = \eta_j} \phi(w_p(\eta_j)|c_n|).$$

After summing over the distinct locations, the measure objective is no larger than the displayed finite-network objective. Thus the measure minimum is no larger than the right-hand infimum in (27). The two infima are equal, and the finite atom representation of a global minimizer attains the right-hand side, so both infima are minima.

Suppose $V \neq 0$. Then a global minimizer μ_* has positive objective, because Theorem 3.20 would otherwise give $\mu_* = 0$, whose fidelity is positive. By comparison with the zero measure,

$$\begin{aligned}
J(\mu_*) &\leq J(0) = \frac{1}{2} \|V\|_{H^1(D)}^2 = \alpha\phi(T_0), \\
\|r_{\mu_*}\|_{H^1(D)} &\leq \|V\|_{H^1(D)}, && \text{since } L(\mu_*) \leq J(\mu_*).
\end{aligned}$$

Hence

$$(31) \quad T(\mu_*) \leq T_0, \quad \gamma_{T(\mu_*)} \geq \gamma_{T_0}.$$

For an atom ω_i of normalized magnitude $t_i = |\mu_{*,p}(\{\omega_i\})|$, Lemma 3.19 gives $t_i \leq T(\mu_*)$. Theorem 3.17 then gives

$$\begin{aligned} \alpha\phi'(T_0) &\leq \alpha\phi'(t_i) = |\langle r_{\mu_*}, K_p(\omega_i) \rangle_{H^1(D)}| && \text{since } t_i \leq T(\mu_*) \leq T_0 \\ &\leq \|r_{\mu_*}\|_{H^1(D)} \|K_p(\omega_i)\|_{H^1(D)} && \text{by Cauchy-Schwarz} \\ &\leq \|V\|_{H^1(D)} \|K_p(\omega_i)\|_{H^1(D)}. \end{aligned}$$

Thus every atom belongs to $\mathcal{K}_0$. The level in (29) is strictly positive, while $K_p \in C_0(\Omega; H^1(D))$, so $\mathcal{K}_0$ is compact. Combining (31) with (24) gives pairwise separation at least $\sqrt{2\alpha\gamma_{T_0}}$. The covering argument from Theorem 3.20 on $K_p(\mathcal{K}_0)$ proves (30).

If $V = 0$, then $J(0) = 0$. Nonnegativity makes zero the minimum, and the zero-objective conclusion of Theorem 3.20 shows that no other global minimizer exists. $\square$

The hypotheses used so far are jointly satisfiable, and the following family makes the constants explicit.

Example 3.22 (separation fails for convex penalty). For $\gamma \geq 0$, define

$$(32) \quad \phi_\gamma(z) := \begin{cases} \frac{z}{2} + \frac{1}{4\gamma} \log(1 + 2\gamma z), & \gamma > 0, \\ z, & \gamma = 0. \end{cases}$$

Then $\phi_\gamma(0) = 0$ and, for $\gamma > 0$,

$$\phi'_\gamma(z) = \frac{1}{2} + \frac{1}{2(1 + 2\gamma z)} \in \left(\frac{1}{2}, 1\right], \quad \phi''_\gamma(t) = -\frac{\gamma}{(1 + 2\gamma t)^2}.$$

Hence $L_{\phi_\gamma} = \phi'_\gamma(0) = 1 \in (0, \infty)$, which is Assumption 3.2; the derivative is positive and nonincreasing and $\phi_\gamma(z) \geq z/2 \rightarrow \infty$, which is Assumption 3.3. For Assumption 3.4, the second derivative is continuous and negative on $(0, \infty)$, part (1) holds with $\gamma_1 = \gamma$ and any $\hat{z}_1 > 0$, and part (2) holds with

$$(33) \quad \gamma_A = \frac{\gamma}{(1 + 2\gamma A)^2} > 0.$$

Thus ϕ_γ with $\gamma > 0$ satisfies Assumptions 3.2–3.4, and (33) turns the atom separation $\sqrt{2\alpha\gamma_{T_0}}$ and the neuron bound (30) into computable quantities. This is the penalty used by Algorithm 1.

The limit $\gamma \rightarrow 0$ is the convex endpoint. Assumption 3.4 fails at $\gamma = 0$, so Theorem 3.20 and Corollary 3.21 do not apply; the weaker conclusions of Theorems 3.16, 3.17, and 5.1 remain available. The degeneration is quantitative rather than abrupt: by (33), $0 < \gamma_{T_0} \leq \gamma \rightarrow 0$ as $\gamma \rightarrow 0$. Thus the guaranteed separation scale tends to zero; the corresponding covering-number bound may deteriorate or become uninformative.

For fixed, distinct locations $\omega = (\omega_n)_{n=1}^N \in \Omega^N$, define the reduced coefficient objective

$$(34) \quad J_\omega(c) := L \left(\sum_{n=1}^N c_n \delta_{\omega_n} \right) + \alpha \sum_{n=1}^N \phi(w_p(\omega_n) |c_n|), \quad c \in \mathbb{R}^N.$$

Theorem 3.23 (Sufficient condition for local optimality). *Let $\phi : [0, \infty) \rightarrow [0, \infty)$ satisfy $\phi(0) = 0$ and $\phi \in C^1((0, \infty))$, with $0 < L_\phi := \phi'(0+) < \infty$. Suppose there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that*

$$(35) \quad \phi(z) \geq L_\phi z - \frac{\gamma_1}{2} z^2 \quad (0 \leq z \leq \hat{z}_1).$$

Let $p > s_1$, let $V \in H^1(D)$, and suppose that Assumption 3.1 holds. Let

$$\bar{\omega} := (\bar{\omega}_n)_{n=1}^N, \quad \bar{c} := (\bar{c}_n)_{n=1}^N \in (\mathbb{R} \setminus \{0\})^N, \quad \bar{\mu} := \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n},$$

where the points $\bar{\omega}_n$ are distinct and every $\bar{c}_n$ is nonzero. Suppose also that

$$(36) \quad |\phi'(w_p(\bar{\omega}_n)|\bar{c}_n|)| < L_\phi \quad (n = 1, \dots, N).$$

Define

$$\bar{P}_p(\omega) := \langle \mathcal{N}\bar{\mu} - V, K_p(\omega) \rangle_{H^1(D)}.$$

Assume that

- (1) $\bar{c} \in (\mathbb{R} \setminus \{0\})^N$ is a local minimizer of $J_{\bar{\omega}}$;
- (2) the strict inequality

$$|\bar{P}_p(\omega)| < \alpha L_\phi$$

holds for every $\omega \in \Omega \setminus \{\bar{\omega}_1, \dots, \bar{\omega}_N\}$.

Then $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$ -local minimizer of J .

Proof. We use the normalized coordinate to reduce the claim to total-variation estimates. Set

$$\bar{z} := (\bar{z}_n)_{n=1}^N, \quad \bar{z}_n := w_p(\bar{\omega}_n)\bar{c}_n.$$

For $z = (z_n)_{n=1}^N \in \mathbb{R}^N$, define

$$\hat{J}(z) := \frac{1}{2} \left\| \sum_{n=1}^N z_n K_p(\bar{\omega}_n) - V \right\|_{H^1(D)}^2 + \alpha \sum_{n=1}^N \phi(|z_n|).$$

The invertible diagonal change $z_n = w_p(\bar{\omega}_n)c_n$ shows that $\bar{z}$ is a local minimizer of $\hat{J}$.

We first obtain a uniform strict gap. Stationarity of $J_{\bar{\omega}}$ at $\bar{c}$ gives

$$\bar{P}_p(\bar{\omega}_n) = -\alpha \phi'(|\bar{z}_n|) \text{sign}(\bar{z}_n).$$

The active-slope gap (36) and condition (2) give

$$|\bar{P}_p(\omega)| < \alpha L_\phi \quad (\omega \in \Omega).$$

Since $\bar{P}_p \in C_0(\Omega)$, the positive continuous function $\alpha L_\phi - |\bar{P}_p|$ tends to αL_ϕ at infinity. Hence there is $m > 0$ such that

$$(37) \quad |\bar{P}_p(\omega)| \leq \alpha L_\phi - m \quad (\omega \in \Omega).$$

Choose $\varepsilon_1 > 0$ such that $\hat{J}(z) \geq \hat{J}(\bar{z})$ whenever $\|z - \bar{z}\|_{\ell^1} \leq \varepsilon_1$. Put

$$B_{\bar{\omega}} := \max_{1 \leq n \leq N} \|K_p(\bar{\omega}_n)\|_{H^1(D)}.$$

Choose $\varepsilon > 0$ so that

$$\varepsilon \leq \varepsilon_1, \quad \varepsilon \leq \hat{z}_1, \quad B_p B_{\bar{\omega}} \varepsilon \leq \frac{m}{2}, \quad \frac{\alpha \gamma_1 \varepsilon}{2} \leq \frac{m}{4}.$$

A condition with zero left-hand side is void.

Let $\mu \in \mathcal{M}_p(\Omega)$ satisfy $\|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} \leq \varepsilon$. There is nothing to prove if $J(\mu) = +\infty$. Otherwise, write $\xi := \mu_p$ and, for $1 \leq n \leq N$, set

$$z_n := \xi(\{\bar{\omega}_n\}), \quad z := (z_n)_{n=1}^N, \quad \xi_0 := \sum_{n=1}^N z_n \delta_{\bar{\omega}_n}, \quad \xi = \xi_0 + \tilde{\xi}.$$

Then $\tilde{\xi}$ has no mass at the support points, and the normalization isometry gives

$$\|z - \bar{z}\|_{\ell^1} + \|\tilde{\xi}\|_{\mathcal{M}(\Omega)} = \|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} \leq \varepsilon.$$

Define

$$r' := \sum_{n=1}^N (z_n - \bar{z}_n) K_p(\bar{\omega}_n), \quad Q_p(\omega) := \langle r', K_p(\omega) \rangle_{H^1(D)}.$$

Then

$$\begin{aligned} \|r'\|_{H^1(D)} &\leq B_{\bar{\omega}} \|z - \bar{z}\|_{\ell^1} \leq B_{\bar{\omega}} \varepsilon, \\ |Q_p(\omega)| &\leq B_p B_{\bar{\omega}} \varepsilon \leq \frac{m}{2}. \end{aligned}$$

Combining this with (37) gives

$$(38) \quad |\bar{P}_p(\omega) + Q_p(\omega)| \leq \alpha L_\phi - \frac{m}{2} \quad (\omega \in \Omega).$$

The exact quadratic expansion of the fidelity in the direction $\tilde{\xi}$ and the nonnegativity of its quadratic remainder give

$$(39) \quad L(\mu) \geq \frac{1}{2} \left\| \sum_{n=1}^N z_n K_p(\bar{\omega}_n) - V \right\|_{H^1(D)}^2 + \int_{\Omega} (\bar{P}_p + Q_p) d\tilde{\xi}.$$

The measures ξ_0 and $\tilde{\xi}$ are mutually singular. Every atom of $\tilde{\xi}$ has magnitude at most $\|\tilde{\xi}\|_{\mathcal{M}(\Omega)} \leq \varepsilon \leq \hat{z}_1$. Moreover,

$$\Phi_\phi(\xi) = \sum_{n=1}^N \phi(|z_n|) + \Phi_\phi(\tilde{\xi}).$$

The local lower bound (35), applied to the atoms and combined with the definition on the continuous part, gives the linear term $L_\phi \|\tilde{\xi}\|_{\mathcal{M}(\Omega)}$ and an atomic remainder bounded below by

$$-\frac{\gamma_1}{2} \sum_{\omega \in \text{atom } \tilde{\xi}} |\tilde{\xi}(\{\omega\})|^2 \geq -\frac{\gamma_1 \varepsilon}{2} \sum_{\omega \in \text{atom } \tilde{\xi}} |\tilde{\xi}(\{\omega\})|,$$

because every atom magnitude is at most ε . Therefore

$$(40) \quad \Phi_\phi(\tilde{\xi}) \geq \left(L_\phi - \frac{\gamma_1 \varepsilon}{2} \right) \|\tilde{\xi}\|_{\mathcal{M}(\Omega)}.$$

Using (39), (38), and (40), we obtain

$$\begin{aligned} J(\mu) &\geq \hat{J}(z) + \int_{\Omega} (\bar{P}_p + Q_p) d\tilde{\xi} + \alpha \Phi_\phi(\tilde{\xi}) \\ &\geq \hat{J}(z) + \left(\frac{m}{2} - \frac{\alpha \gamma_1 \varepsilon}{2} \right) \|\tilde{\xi}\|_{\mathcal{M}(\Omega)} \\ &\geq \hat{J}(\bar{z}) + \frac{m}{4} \|\tilde{\xi}\|_{\mathcal{M}(\Omega)} \geq J(\bar{\mu}). \end{aligned}$$

Thus $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$ -local minimizer. $\square$

Admissible activations. The two exponents in Assumption 3.1 play different roles: s_0 controls the value atom in $L^2(D)$ and is enough for Theorem 3.16; s_1 controls the full atom in $H^1(D)$ and is used by Theorems 3.17, 3.20, and 5.1. Since $\|K(\omega)\|_{L^2(D)} \leq \|K(\omega)\|_{H^1(D)}$, it is consistent to impose $s_0 \leq s_1$, as Assumption 3.1 does. The exponent s_1 is obtained by estimating $a\rho'(a \cdot x + b)$ separately. The condition $p > s_1$ normalizes the full H^1 atom; together with Assumption 3.4(2), it is sufficient for the finite-support conclusion in Theorem 3.20. The following exponents satisfy

the pointwise bounds on a bounded domain.

activation	s_0	s_1	sufficient condition
$\tanh z$	0	1	$p > 1$
e^{-z^2}	0	1	$p > 1$
$\log(1 + e^z)$	1	1	$p > 1$
$(z\Phi(z))^2$	2	2	$p > 2$
$(z_+)^k$	k	k	$p > k$

For bounded activations such as $\tanh$ and the Gaussian, existence alone requires only $p > 0$; the condition $p > 1$ in the table is the one that also dominates the elementary global H^1 growth estimate. The fourth row is the squared Gaussian error linear unit, written GELU² below, with Φ the standard normal distribution function; since Φ takes values in $(0, 1)$ we have $|z\Phi(z)| \leq |z|$, and its derivative $2z\Phi(z)(\Phi(z) + z\varphi(z))$ grows linearly, which gives $s_0 = s_1 = 2$. The last row is listed for comparison with the positively homogeneous formulation of Section 4.

4. THE k -HOMOGENEOUS FORMULATION

The ReLU^k activation function is of particular interest since it is widely used in the Sobolev training [14, 11]. For gradient fitting with ReLU^k we use $k \geq 2$, so that $K(\omega) \in C^1(\overline{D})$. As we will see in this section, the induced penalty is $\psi_k(t) = t^q$ with $q = 2/(k+1) < 1$, so it has infinite right slope at zero. It does not extend the result in Section 3.

On the other hand, k -homogeneity makes the radial part of an inner parameter redundant: every atom can be moved onto the unit sphere $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ at the cost of rescaling its outer coefficient. The parameter domain of the resulting problem is compact, so the dictionary is bounded, and neither the polynomial weight w_p nor the weighted space $\mathcal{M}_p(\Omega)$ of Section 3 is required. Throughout this section we work on $\mathcal{M}(\mathbb{S}^d)$ with the total-variation norm $\|\cdot\|_{\mathcal{M}(\mathbb{S}^d)}$, which we identify with the subspace of $\mathcal{M}(\Omega)$ of measures vanishing outside $\mathbb{S}^d$.

Assumption 4.1 (k -homogeneity). For some $k \geq 1$,

$$K(\lambda\omega) = \lambda^k K(\omega) \quad (\lambda > 0, \omega \in \mathbb{R}^{d+1}).$$

Lemma 4.2 (The ReLU^k dictionary). Let $k \geq 2$, let $D \subset \mathbb{R}^d$ be bounded with positive Lebesgue measure, let $\rho(z) = (z_+)^k$, and let

$$[K(\omega)](x) = (a \cdot x + b)_+^k, \quad \omega = (a, b) \in \mathbb{R}^{d+1}.$$

Then

- (1) K satisfies Assumption 4.1 with degree k ;
- (2) $K(\omega) \in C^1(\mathbb{R}^d)$ with $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1}a$, and this classical gradient is also the distributional gradient of $K(\omega)$ on D ;
- (3) $K : \mathbb{S}^d \rightarrow H^1(D)$ is continuous; and
- (4) $C_S := \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}$ satisfies $0 < C_S < \infty$.

Proof. (1) For $\lambda > 0$ one has $(\lambda z)_+ = \lambda z_+$, hence $(\lambda a \cdot x + \lambda b)_+^k = \lambda^k (a \cdot x + b)_+^k$ for every x .

(2) Set $h_k(z) := (z_+)^k$. On $(0, \infty)$ and on $(-\infty, 0)$ the function h_k is differentiable with $h'_k(z) = k(z_+)^{k-1}$. At $z = 0$ both one-sided difference quotients are $O(|z|^{k-1})$ and vanish in the limit because $k \geq 2$, so $h'_k(0) = 0 = k(0_+)^{k-1}$. The formula $h'_k(z) = k(z_+)^{k-1}$ therefore holds on all of $\mathbb{R}$, and h'_k is continuous because $k - 1 \geq 1$. The chain rule applied to $x \mapsto a \cdot x + b$ gives $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1}a$. Since D is bounded, both $K(\omega)$ and $\nabla K(\omega)$ are bounded and continuous on D ; a continuously differentiable function on a neighbourhood of D lies in $H^1(D)$ and its distributional gradient is its classical gradient.

(3) Let $\omega_j = (a_j, b_j) \rightarrow \omega = (a, b)$ in $\mathbb{S}^d$ and put $R_D := \sup_{x \in D} |x| < \infty$. Then

$$\sup_{x \in D} |(a_j \cdot x + b_j) - (a \cdot x + b)| \leq |a_j - a| R_D + |b_j - b| \rightarrow 0,$$

so the pre-activations converge uniformly on D . All of them lie in the compact interval $I := [-(R_D + 1), R_D + 1]$, because $|a_j \cdot x + b_j| \leq R_D + 1$ for every j and every $x \in D$ by $|a_j| \leq 1$, $|b_j| \leq 1$. The functions h_k and h'_k are uniformly continuous on I , whence

$$K(\omega_j) \rightarrow K(\omega) \quad \text{and} \quad h'_k(a_j \cdot x + b_j) \rightarrow h'_k(a \cdot x + b) \quad \text{uniformly on } D.$$

Since $\nabla K(\omega_j)(x) = h'_k(a_j \cdot x + b_j) a_j$, $a_j \rightarrow a$, and the scalar factors are uniformly bounded on D , the gradients converge uniformly on D as well. As D has finite Lebesgue measure, uniform convergence implies convergence in $L^2(D)$, so $K(\omega_j) \rightarrow K(\omega)$ in $H^1(D)$.

(4) Finiteness follows from (3) and compactness of $\mathbb{S}^d$, since a continuous real function attains a finite maximum on a compact set. For positivity take $\hat{\omega} = (0, 1) \in \mathbb{S}^d$, for which $[K(\hat{\omega})](x) = 1$ for every x ; hence $C_S \geq \|K(\hat{\omega})\|_{H^1(D)} = |D|^{1/2} > 0$. $\square$

In the remainder of the section $K : \mathbb{S}^d \rightarrow H^1(D)$ is continuous and Lemma 4.2 verifies this assumption and $C_S > 0$ for ReLU^k with $k \geq 2$, the range used for the gradient-augmented problem.

The following lemma extends the degree-one ReLU rescaling result [20, Appendix F, Proposition 3] to a k -homogeneous dictionary. We retain general positive exponents to make explicit how the homogeneity degree changes both the induced exponent and its coefficient.

Lemma 4.3 (Rescaling of a k -homogeneous network). *Let Assumption 4.1 hold, and let $\xi, \zeta > 0$. For a fixed nonzero represented coefficient $\tilde{c}$ and a unit direction $\hat{\omega} \in \mathbb{S}^d$ with $K(\hat{\omega}) \neq 0$, minimize*

$$\frac{|c|^\xi}{\xi} + \frac{|\omega|^\zeta}{\zeta}$$

over radial representations

$$\omega = \lambda \hat{\omega}, \quad \lambda > 0, \quad c K(\omega) = \tilde{c} K(\hat{\omega}).$$

The minimum equals

$$(41) \quad C_{k,\xi,\zeta} |\tilde{c}|^q, \quad q := \frac{\xi\zeta}{k\xi + \zeta}, \quad C_{k,\xi,\zeta} := \frac{1}{q} k^{-k\xi/(k\xi + \zeta)}.$$

Consequently, consider any finite representation without zero-function neurons and set

$$\hat{\omega}_n := \frac{\omega_n}{|\omega_n|}, \quad \tilde{c}_n := c_n |\omega_n|^k.$$

Among all independent radial rescalings that preserve each function $c_n K(\omega_n)$, the minimum total penalty is $C_{k,\xi,\zeta} \sum_n |\tilde{c}_n|^q$.

Proof. We reduce the rescaling constraint to a scalar minimization and solve it explicitly. Write $\omega = \lambda \hat{\omega}$, where $\lambda = |\omega| > 0$. Since $K(\hat{\omega}) \neq 0$, k -homogeneity makes the representation constraint equivalent to $c \lambda^k = \tilde{c}$. Hence, for fixed $\tilde{c}$, the radial penalty is

$$g(\lambda) = \frac{|\tilde{c}|^\xi}{\xi \lambda^{k\xi}} + \frac{\lambda^\zeta}{\zeta}.$$

It tends to $+\infty$ at both endpoints, and

$$g'(\lambda) = \lambda^{-k\xi-1} \left(\lambda^{k\xi+\zeta} - k |\tilde{c}|^\xi \right).$$

Thus g' changes sign exactly once, at

$$\lambda_* = (k |\tilde{c}|^\xi)^{1/(k\xi + \zeta)}.$$

It is therefore the unique global minimizer. The stationarity identity gives $|\tilde{c}|^\xi \lambda_*^{-k\xi} = \lambda_*^\zeta/k$; using the definition of q in (41), we obtain

$$\begin{aligned} g(\lambda_*) &= \left(\frac{1}{k\xi} + \frac{1}{\zeta} \right) \lambda_*^\zeta \\ &= \frac{1}{q} k^{-k\xi/(k\xi+\zeta)} |\tilde{c}|^q \\ &= C_{k,\xi,\zeta} |\tilde{c}|^q. \end{aligned}$$

For each normalized neuron $(\tilde{c}_n, \hat{\omega}_n)$, set

$$\lambda_n = (k|\tilde{c}_n|^\xi)^{1/(k\xi+\zeta)}, \quad \omega_n = \lambda_n \hat{\omega}_n, \quad c_n = \tilde{c}_n \lambda_n^{-k}.$$

By k -homogeneity, $c_n K(\omega_n) = \tilde{c}_n K(\hat{\omega}_n)$, and the displayed rescaling preserves this function. The radial variables are independent, so applying the scalar minimum to each neuron and summing proves the final assertion. $\square$

Remark 4.4 (The quadratic case and the meaning of α). For $\xi = \zeta = 2$,

$$q = \frac{2}{k+1}, \quad C_k := C_{k,2,2} = \frac{k+1}{2} k^{-k/(k+1)}.$$

In particular $C_1 = 1$ and $C_2 = \frac{3}{2} 2^{-2/3}$. Since C_k depends on k , equal nominal values of α in the penalty $\alpha \sum_n |c_n|^q$ for different powers k do *not* correspond to equal coefficients of the original separable penalty.

For $k \geq 2$ define the scalar penalty

$$\psi_k(t) := t^q, \quad q := \frac{2}{k+1} \in (0, 1), \quad t \geq 0,$$

which is the exponent of Lemma 4.3 for $\xi = \zeta = 2$. It is continuous and strictly increasing with $\psi_k(0) = 0$ and $\psi'_k(0^+) = +\infty$, and it is subadditive:

$$(42) \quad \psi_k(s+t) \leq \psi_k(s) + \psi_k(t) \quad (s, t \geq 0).$$

Indeed, for $s+t > 0$ the numbers $u := s/(s+t)$ and $1-u = t/(s+t)$ lie in $[0, 1]$, where $v^q \geq v$; hence $\psi_k(s) + \psi_k(t) = (s+t)^q (u^q + (1-u)^q) \geq (s+t)^q$. Iterating (42) and passing to the limit in the finite partial sums gives, for every summable sequence $(t_j)_{j \geq 1}$ of nonnegative numbers,

$$(43) \quad \psi_k\left(\sum_{j \geq 1} t_j\right) \leq \sum_{j \geq 1} \psi_k(t_j).$$

Definition 4.5 (Homogeneous measure penalty). For $\mu \in \mathcal{M}(\mathbb{S}^d)$ with atomic and continuous parts μ_{atom} and μ_{cont} as in Section 2.1, the atom set is at most countable. Hence the following sum of nonnegative terms is well defined in $[0, +\infty]$ and independent of its enumeration. Set

$$\Phi_{\psi_k}(\mu) := \begin{cases} \sum_{\omega \in \text{atom } \mu} \psi_k(|\mu(\{\omega\})|), & \mu_{\text{cont}} = 0, \\ +\infty, & \mu_{\text{cont}} \neq 0. \end{cases}$$

The functional is nonnegative, vanishes only at the zero measure, and, whenever $\Phi_{\psi_k}(\mu) < \infty$, countable subadditivity gives

$$(44) \quad \Phi_{\psi_k}(\mu) \geq \psi_k(\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}) = \|\mu\|_{\mathcal{M}(\mathbb{S}^d)}^q.$$

For a finite representation $\mu = \sum_{n=1}^N c_n \delta_{\omega_n}$, repeated locations are merged. The triangle inequality and (42) show that merging cannot increase $\sum_n \psi_k(|c_n|)$.

4.1. **The measure problem on the sphere.** For $\mu \in \mathcal{M}(\mathbb{S}^d)$ we now set

$$(45) \quad J_{\psi_k}(\mu) := L(\mu) + \alpha \Phi_{\psi_k}(\mu),$$

$$(46) \quad J_{\psi_k}^M(\mu) := l^M(\mu) + \alpha \Phi_{\psi_k}(\mu),$$

with the convention that a sum containing $+\infty$ equals $+\infty$. Since the empirical fidelity is also one half of a squared Hilbert-space norm and its dictionary is continuous on the compact sphere, all results below hold for the empirical objective as well. The empirical fidelity (6) was introduced for finite networks; for ReLU^k with $k \geq 2$ it is defined on all of $\{\mu \in \mathcal{M}(\mathbb{S}^d) : \Phi_{\psi_k}(\mu) < \infty\}$, provided the sampled data $V(x^m)$ and g^m are given. Indeed, such a μ is purely atomic with $\sum_j |c_j| < \infty$, while Lemma 4.2(2)–(3) and compactness of $\mathbb{S}^d$ bound $|[K(\omega)](x^m)|$ and $|\nabla K(\omega)(x^m)|$ uniformly in $\omega \in \mathbb{S}^d$ and m ; hence $\sum_j c_j [K(\omega_j)](x^m)$ and $\sum_j c_j \nabla K(\omega_j)(x^m)$ converge absolutely, so both residuals in $l^M(\mu)$ are defined. We study the minimization problem

$$(P_{\psi_k}) \quad \min_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu).$$

For a finite network with distinct locations $\omega_n \in \mathbb{S}^d$, the definition of Φ_{ψ_k} gives

$$J_{\psi_k}(\mu_{\omega,c}) = L(\mu_{\omega,c}) + \alpha \sum_{n=1}^N |c_n|^q,$$

which is the objective minimized by the finite-dimensional solver in Algorithm 2; the empirical objective (46) restricts in the same way.

Local minimizers of (P_{ψ_k}) are understood as in Definition 3.14, with $\mathcal{M}_p(\Omega)$ replaced by $\mathcal{M}(\mathbb{S}^d)$, J by J_{ψ_k} , and the weighted measure norm by the total-variation norm.

Proposition 4.6 (Finite truncations and equality of the infima). *Let $V \in H^1(D)$, let $K : \mathbb{S}^d \rightarrow H^1(D)$ be continuous with bound C_S , let $\mu \in \mathcal{M}(\mathbb{S}^d)$ satisfy $\Phi_{\psi_k}(\mu) < \infty$, let $(\omega_j)_{j \in \mathcal{I}}$, $\mathcal{I} \subseteq \mathbb{N}$, enumerate atom μ , write $c_j := \mu(\{\omega_j\}) \neq 0$, and set*

$$\mu_m := \sum_{j \in \mathcal{I}, j \leq m} c_j \delta_{\omega_j}.$$

Then

$$\begin{aligned} \mathcal{N}\mu &\in H^1(D), \quad \|\mathcal{N}\mu\|_{H^1(D)} \leq C_S \|\mu\|_{\mathcal{M}(\mathbb{S}^d)}, \\ \|\mathcal{N}\mu_m - \mathcal{N}\mu\|_{H^1(D)} &\leq C_S \|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} \longrightarrow 0. \end{aligned}$$

Moreover, $J_{\psi_k}(\mu_m) \rightarrow J_{\psi_k}(\mu)$. Consequently the all-measures infimum and the finite-network infimum coincide:

$$(47) \quad \inf_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu) = \inf_{N \in \mathbb{N}_0} \inf_{\substack{\omega_n \in \mathbb{S}^d, \\ n=1, \dots, N}} \inf_{c_n \in \mathbb{R}} \left[L(\mu_{\omega,c}) + \alpha \sum_{n=1}^N \psi_k(|c_n|) \right].$$

Proof. Since $\Phi_{\psi_k}(\mu) < \infty$, Definition 4.5 gives $\mu_{\text{cont}} = 0$, so $\mu = \sum_{j \in \mathcal{I}} c_j \delta_{\omega_j}$ with distinct ω_j and $c_j \neq 0$. Since $\sum_{j \in \mathcal{I}} |c_j| = \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} < \infty$, the tail identity

$$\|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} = \sum_{j \in \mathcal{I}, j > m} |c_j| \longrightarrow 0.$$

Bounded linearity of $\mathcal{N}$ therefore gives $\mathcal{N}\mu \in H^1(D)$ and the stated norm bounds. Continuity of $u \mapsto \frac{1}{2} \|u - V\|_{H^1(D)}^2$ gives $L(\mu_m) \rightarrow L(\mu)$, while

$$\Phi_{\psi_k}(\mu_m) = \sum_{j \in \mathcal{I}, j \leq m} \psi_k(|c_j|) \longrightarrow \Phi_{\psi_k}(\mu)$$

by monotone convergence. Combining these two limits proves $J_{\psi_k}(\mu_m) \rightarrow J_{\psi_k}(\mu)$.

It remains to compare the infima; call them i_{meas} and i_{disc} , in the order in which they appear in (47). Every finite network defines an admissible measure, and its finite-dimensional objective

equals the corresponding measure objective. Hence $i_{\text{meas}} \leq i_{\text{disc}}$. Conversely, let μ satisfy $J_{\psi_k}(\mu) < \infty$. Its truncations μ_m are finite networks and satisfy $J_{\psi_k}(\mu_m) \rightarrow J_{\psi_k}(\mu)$. Therefore $i_{\text{disc}} \leq J_{\psi_k}(\mu)$. Taking the infimum over such μ gives $i_{\text{disc}} \leq i_{\text{meas}}$. $\square$

Proposition 4.6 shows that a measure of finite penalty automatically has finite fidelity. Since conversely $J_{\psi_k}(\mu) < \infty$ contains $\alpha\Phi_{\psi_k}(\mu) < \infty$, the two conditions agree:

$$J_{\psi_k}(\mu) < \infty \iff \Phi_{\psi_k}(\mu) < \infty \quad (\mu \in \mathcal{M}(\mathbb{S}^d)),$$

and either may be used as the standing finiteness hypothesis.

Proposition 4.6 is used only as a bridge between the finite-network problem and (P_{ψ_k}) . It equates the two infima; attainment is established in Theorem 4.9 from the finite-dimensional argument and is not claimed here.

Theorem 4.7. *Let $k \geq 2$, $q = 2/(k+1) \in (0, 1)$, $\alpha > 0$, $V \in H^1(D)$, and let $K : \mathbb{S}^d \rightarrow H^1(D)$ be continuous. Let $\bar{\mu} \in \mathcal{M}(\mathbb{S}^d)$ satisfy $J_{\psi_k}(\bar{\mu}) < \infty$ and be optimal against perturbations of the weights of its own atoms: there is $\varepsilon > 0$ such that*

$$(48) \quad J_{\psi_k}(\bar{\mu}) \leq J_{\psi_k}(\bar{\mu} + t\delta_\omega) \quad \text{for every } \omega \in \text{atom } \bar{\mu} \text{ and every } |t| < \varepsilon.$$

Set

$$B(\bar{\mu}) := \|P_{\bar{\mu}}\|_{C(\mathbb{S}^d)}.$$

Then $\bar{\mu}$ is purely atomic, $B(\bar{\mu}) < \infty$, and every atom $\omega_n \in \text{atom } \bar{\mu}$ with weight $c_n := \bar{\mu}(\{\omega_n\})$ satisfies

$$(49) \quad P_{\bar{\mu}}(\omega_n) + \alpha q |c_n|^{q-1} \text{sign}(c_n) = 0.$$

If $B(\bar{\mu}) = 0$, then $\bar{\mu} = 0$. If $B(\bar{\mu}) > 0$, then

$$(50) \quad |c_n| \geq c_{\min}(\bar{\mu}) := \left(\frac{\alpha q}{B(\bar{\mu})} \right)^{1/(1-q)} > 0 \quad \text{for every } \omega_n \in \text{atom } \bar{\mu}.$$

Hypothesis (48) holds in particular if $\bar{\mu}$ is a local minimizer of (P_{ψ_k}) in total variation in the sense specified above.

Proof. We derive (49) from a two-sided perturbation of a single atom weight, and then convert it into the lower bound (50).

Since $J_{\psi_k}(\bar{\mu}) < \infty$ we have $\Phi_{\psi_k}(\bar{\mu}) < \infty$, so $\bar{\mu}_{\text{cont}} = 0$ and $\bar{\mu}$ is purely atomic. By Proposition 4.6, applied to $\bar{\mu}$, $\mathcal{N}\bar{\mu} \in H^1(D)$, so $r_{\bar{\mu}} \in H^1(D)$ is well defined. The map $P_{\bar{\mu}}$ is continuous on $\mathbb{S}^d$, being the composition of the continuous map K with the continuous linear functional $\langle r_{\bar{\mu}}, \cdot \rangle_{H^1(D)}$, and $\mathbb{S}^d$ is compact, so $B(\bar{\mu}) < \infty$.

Fix $\omega_n \in \text{atom } \bar{\mu}$, write $c_n = \bar{\mu}(\{\omega_n\}) \neq 0$, let $s \in \{-1, 1\}$, and consider $\mu_t := \bar{\mu} + ts\delta_{\omega_n}$ for $0 < t < \min\{\varepsilon, |c_n|\}$. The measure μ_t differs from $\bar{\mu}$ only in the weight of the atom ω_n , which becomes $c_n + ts \neq 0$, so $\text{atom } \mu_t = \text{atom } \bar{\mu}$ and

$$\Phi_{\psi_k}(\mu_t) - \Phi_{\psi_k}(\bar{\mu}) = \psi_k(|c_n + ts|) - \psi_k(|c_n|),$$

all other summands being unchanged. For the fidelity, $\mathcal{N}\mu_t = \mathcal{N}\bar{\mu} + tsK(\omega_n)$, so the residual is $r_{\bar{\mu}} + tsK(\omega_n)$ and expanding the square gives the exact identity

$$\begin{aligned} L(\mu_t) - L(\bar{\mu}) &= ts \langle r_{\bar{\mu}}, K(\omega_n) \rangle_{H^1(D)} + \frac{t^2}{2} \|K(\omega_n)\|_{H^1(D)}^2 \\ &= ts P_{\bar{\mu}}(\omega_n) + \frac{t^2}{2} \|K(\omega_n)\|_{H^1(D)}^2. \end{aligned}$$

Because $c_n \neq 0$, the map $z \mapsto |z|^q$ is differentiable at c_n with derivative $q|c_n|^{q-1} \text{sign}(c_n)$. Hypothesis (48) gives $J_{\psi_k}(\mu_t) - J_{\psi_k}(\bar{\mu}) \geq 0$; dividing by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \leq s \left[P_{\bar{\mu}}(\omega_n) + \alpha q |c_n|^{q-1} \text{sign}(c_n) \right].$$

Choosing $s = 1$ and $s = -1$ gives (49).

Suppose $B(\bar{\mu}) = 0$, that is, $P_{\bar{\mu}} \equiv 0$ on $\mathbb{S}^d$. If $\bar{\mu}$ had an atom ω_n , then (49) would read $\alpha q |c_n|^{q-1} \text{sign}(c_n) = 0$, which is impossible because $\alpha > 0$, $q > 0$, and $|c_n|^{q-1} \in (0, \infty)$ for $c_n \neq 0$. Hence atom $\bar{\mu} = \emptyset$, and since $\bar{\mu}$ is purely atomic, $\bar{\mu} = 0$.

Suppose $B(\bar{\mu}) > 0$. Taking absolute values in (49) gives

$$\alpha q |c_n|^{q-1} = |P_{\bar{\mu}}(\omega_n)| \leq B(\bar{\mu}).$$

Since $q - 1 = -(1 - q) < 0$, this reads $|c_n|^{-(1-q)} \leq B(\bar{\mu})/(\alpha q)$, hence $|c_n|^{1-q} \geq \alpha q/B(\bar{\mu})$; raising both sides to the positive power $1/(1 - q)$ preserves the inequality and gives (50). $\square$

Remark 4.8. The identity $\psi'_k(0^+) = +\infty$ has the following consequences.

- (1) Inserting a new atom at $\omega \notin \text{atom } \bar{\mu}$ with weight ts , $s \in \{-1, 1\}$, changes the penalty by αt^q . Since $t^q/t = t^{q-1} \rightarrow \infty$ as $t \downarrow 0$, division by t produces no finite limit, and hence no pointwise condition on $P_{\bar{\mu}}(\omega)$ away from atom $\bar{\mu}$. Theorem 4.7 is restricted to existing atoms for this reason.
- (2) The zero measure is a local minimizer of (P_{ψ_k}) in total variation. Indeed, $J_{\psi_k}(\mu) = \infty$ whenever $\Phi_{\psi_k}(\mu) = \infty$, while for μ with $\Phi_{\psi_k}(\mu) < \infty$ and $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} = \varepsilon$,

$$\begin{aligned} J_{\psi_k}(\mu) - J_{\psi_k}(0) &= \langle -V, \mathcal{N}\mu \rangle_{H^1(D)} + \frac{1}{2} \|\mathcal{N}\mu\|_{H^1(D)}^2 + \alpha \Phi_{\psi_k}(\mu) \\ &\geq -\|V\|_{H^1(D)} C_S \varepsilon + \alpha \varepsilon^q, \end{aligned}$$

by (44), the bound $\|\mathcal{N}\mu\|_{H^1(D)} \leq C_S \varepsilon$ of Proposition 4.6, and the Cauchy–Schwarz inequality. Since $q < 1$, the right-hand side is positive for every sufficiently small $\varepsilon > 0$.

An insertion criterion based on an infinitesimal weight perturbation is therefore unavailable for this penalty, and Algorithm 2 uses a finite-step decrease test instead.

4.2. Global existence and a uniform width bound.

Theorem 4.9 (Existence and width bound). *Let $k \geq 2$, $q = 2/(k + 1) \in (0, 1)$, $\alpha > 0$, let $V \in H^1(D)$ with $V \neq 0$, and let $K : \mathbb{S}^d \rightarrow H^1(D)$ be continuous with $C_S = \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)} > 0$; the latter holds for ReLU^k by Lemma 4.2(4). Then (P_{ψ_k}) admits a global minimizer, and its minimum value equals the finite-network infimum on the right-hand side of (47). Every global minimizer $\bar{\mu}$ is a finite atomic measure satisfying*

$$\# \text{ atom } \bar{\mu} \leq N_{\max} := \frac{\|V\|_{H^1(D)}^2}{2\alpha c_{\min, \text{glob}}^q}, \quad c_{\min, \text{glob}} := \left(\frac{\alpha q}{\|V\|_{H^1(D)} C_S} \right)^{1/(1-q)}.$$

Remark 4.10. The excluded cases are immediate. If $V = 0$, then $\mu = 0$ is the unique global minimizer by the nonnegativity of the fidelity and the strict positivity of Φ_{ψ_k} away from zero. If $C_S = 0$, then $K \equiv 0$, so the fidelity is constant and the same penalty argument again makes $\mu = 0$ the unique global minimizer. Thus the atom bound is zero in both cases. The assumptions $V \neq 0$ and $C_S > 0$ are needed only to ensure that $c_{\min, \text{glob}} \in (0, \infty)$ and $N_{\max} < \infty$ are well defined.

Proof. We first obtain minimizers at each fixed width by compactness, then derive one width estimate for both the existence argument and arbitrary global minimizers.

For $N \in \mathbb{N}_0$, set

$$F_N(\omega, c) := L(\mu_{\omega, c}) + \alpha \sum_{n=1}^N |c_n|^q, \quad m_N := \inf F_N,$$

with $F_0 = m_0 = \frac{1}{2} \|V\|_{H^1(D)}^2$. For $N \geq 1$, continuity of K , scalar multiplication, and finite addition show that $(\omega, c) \mapsto \sum_{n=1}^N c_n K(\omega_n)$ is continuous into $H^1(D)$; hence F_N is continuous.

Any minimizing sequence may be chosen in the sublevel $F_N \leq m_0 + 1$. Since the fidelity is nonnegative, this sublevel satisfies

$$|c_n| \leq R := \left(\frac{m_0 + 1}{\alpha} \right)^{1/q} \quad (n = 1, \dots, N).$$

It therefore lies in the compact set $(\mathbb{S}^d)^N \times [-R, R]^N$. Compactness gives a convergent subsequence, and continuity shows that its limit attains m_N . The case $N = 0$ is immediate, and we have $m_{N+1} \leq m_N$.

We use the following estimate twice. Suppose that a measure μ satisfies $J_{\psi_k}(\mu) \leq m_0$ and the atomwise optimality hypothesis (48). Theorem 4.7 shows that μ is purely atomic. The nonnegativity of the penalty gives $\|r_\mu\|_{H^1(D)} \leq \|V\|_{H^1(D)}$, and hence

$$B(\mu) \leq \|V\|_{H^1(D)} C_S.$$

If $\mu \neq 0$, Theorem 4.7 gives $B(\mu) > 0$ and

$$|\mu(\{\omega\})| \geq \left(\frac{\alpha q}{B(\mu)} \right)^{1/(1-q)} \geq c_{\min, \text{glob}} \quad (\omega \in \text{atom } \mu).$$

Consequently, for every finite $A \subseteq \text{atom } \mu$,

$$\#A \alpha c_{\min, \text{glob}}^q \leq \alpha \sum_{\omega \in A} |\mu(\{\omega\})|^q \leq \alpha \Phi_{\psi_k}(\mu) \leq J_{\psi_k}(\mu) \leq \frac{1}{2} \|V\|_{H^1(D)}^2.$$

An infinite atom set would contain finite subsets of arbitrarily large cardinality. Thus $\text{atom } \mu$ is finite and its cardinality is at most $N_{\max}$; the same conclusion is immediate when $\mu = 0$.

Set $N_{\text{cut}} := \lfloor N_{\max} \rfloor$. The estimate above applies to every minimizer defining m_N , since each coefficient can be varied independently. Thus, for $N \geq N_{\text{cut}}$, that minimizer has width at most N_{cut} , so $m_{N_{\text{cut}}} \leq m_N$. The reverse inequality follows from the monotonicity of (m_N) . Hence

$$\inf_{N \in \mathbb{N}_0} m_N = m_{N_{\text{cut}}}.$$

Since $m_{N_{\text{cut}}}$ is attained, Proposition 4.6 gives a global minimizer of (P_{ψ_k}) with the same value. Every global minimizer satisfies the hypotheses of the estimate above and therefore has at most $N_{\max}$ atoms. $\square$

Corollary 4.11 (Finite support of total-variation local minimizers). *Let the hypotheses of Theorem 4.7 on k, q, α, V, K hold and let $\bar{\mu} \neq 0$ be a local minimizer of (P_{ψ_k}) in total variation. Then $B(\bar{\mu}) > 0$, the measure $\bar{\mu}$ is finite atomic, and*

$$\# \text{atom } \bar{\mu} \leq \frac{J_{\psi_k}(\bar{\mu})}{\alpha c_{\min}(\bar{\mu})^q}.$$

Proof. By Theorem 4.7, $B(\bar{\mu}) = 0$ would force $\bar{\mu} = 0$; hence $B(\bar{\mu}) > 0$ and every atom weight satisfies $|c_n| \geq c_{\min}(\bar{\mu}) > 0$. Since ψ_k is nondecreasing and the fidelity is nonnegative,

$$N \alpha c_{\min}(\bar{\mu})^q \leq \alpha \Phi_{\psi_k}(\bar{\mu}) \leq J_{\psi_k}(\bar{\mu}) < \infty$$

with $N = \# \text{atom } \bar{\mu} \in \mathbb{N}_0 \cup \{\infty\}$, which forces $N < \infty$ and gives the stated bound. $\square$

The estimate of Corollary 4.11 depends on the minimizer through both $J_{\psi_k}(\bar{\mu})$ and $B(\bar{\mu})$ and is therefore an a posteriori bound. It is not computable in advance, unlike the a priori bound $N_{\max}$ of Theorem 4.9, which is uniform over all global minimizers and computable from $\|V\|_{H^1(D)}$, C_S , α and q alone. On a global minimizer the two are comparable and the a posteriori bound is the sharper one, since $J_{\psi_k}(\bar{\mu}) \leq \frac{1}{2} \|V\|_{H^1(D)}^2$ and $c_{\min}(\bar{\mu}) \geq c_{\min, \text{glob}}$ there.

5. ALGORITHMS

This section describes the computational pipeline: the training data are generated via the open-loop problem, candidate neurons are inserted greedily, and different insertion methods are chosen given the problem setting.

Data generation via the open-loop problem. Set $U := L^2(t_0, T; \mathbb{R}^{m_u})$ and assume that for every $u \in U$ the state equation in (P) has a unique solution $y(u)$ on $[t_0, T]$ and that the control-to-state map $u \mapsto y(u)$ is continuously Fréchet differentiable from U into $C([t_0, T]; \mathbb{R}^d)$. The objective of (P) is then the reduced functional $u \mapsto \mathcal{C}(u; t_0, x)$ on U , and the standard adjoint (Lagrangian) calculus — see, e.g., [12] — gives its gradient

$$\nabla_u \mathcal{C}(u; t_0, x) = 2\eta u + g(y(u))^\top p \quad \text{in } U,$$

where p solves the costate equation of (TPBVP) backward in time along $(y(u), u)$ in place of (y^*, u^*) . Each gradient evaluation therefore costs one forward solve of the state equation and one backward solve of the costate equation, and the implicit derivative of $u \mapsto y(u)$ is never formed. For the data-driven approach it is essential that the initial dataset $\{x^j, V(0, x^j), g^j\}_{j=1}^M$ is *causality-free*: the samples are generated without approximate values at neighboring points. [13] surveys generation methods. As in [1], we minimize $\mathcal{C}(\cdot; t_0, x)$ with Barzilai–Borwein step sizes. The converged initial costate supplies g^j , subject to that reservation and the differentiability qualification of Section 1.

The two algorithms. We use two algorithms of primal–dual active point (PDAP) type, in the sense of the accelerated conditional-gradient method for measure spaces of [21]. Algorithm 1 follows the nonconvex variant of [20], whose scalar penalty satisfies $\phi'(0) < \infty$ as in Assumption 3.2; Algorithm 2 is an adaptation of it to the penalty ψ_k , whose infinite slope at the origin is explicitly excluded by that reference.

In both algorithms the coefficient-correction step freezes the locations ω and applies a semismooth Newton method to the resulting finite-dimensional coefficient problem. They are specified in their respective subsections below. In either case the method is local and its output is retained only when it does not increase the post-insertion objective.

For the sampled fidelity, write $P_\mu^M(\omega) := Dl^M(\mu)[\delta_\omega]$, and let $\|K(\omega)\|_M$ denote the sampled value–gradient norm associated with l^M .

For a joint insertion $\sum_{i=1}^r c_i \delta_{\omega_i}$ at distinct new locations the penalty is additive, but the fidelity change contains pairwise cross terms between the inserted features, which no individual test controls. For $N_{\text{ins}} = 1$ the individual decrease statement applies to every accepted candidate, while Corollary 5.2 additionally requires an exact global maximizer. For $N_{\text{ins}} > 1$ the individual estimates do not compose, and neither a decrease per outer iteration nor monotonicity of the objective along the iterates is claimed. Restoring a batch decrease guarantee requires recomputing the residual after each insertion, or testing the joint increment of the whole batch.

5.1. Algorithm 1: the log penalty on the normalized measure.

Theorem 5.1. *Suppose that Assumption 3.1 holds, $p > s_1$, and $\alpha > 0$. Let ϕ satisfy Assumptions 3.2 and 3.3. For $\mu \in \mathcal{M}_p(\Omega)$ define*

$$\begin{aligned} \Delta(\mu, \omega) &:= \max\{|P_{p,\mu}(\omega)| - \alpha L_\phi, 0\}, \\ \Delta(\mu) &:= \sup_{\omega \in \Omega} \Delta(\mu, \omega) = \max\{\|P_{p,\mu}\|_\infty - \alpha L_\phi, 0\}. \end{aligned}$$

For a finite atomic μ , define

$$(51) \quad R(\mu) := \max \left\{ 1, \left(\frac{2^{s_1+1} C_K \|r_\mu\|_{H^1(D)}}{\alpha L_\phi} \right)^{1/(p-s_1)} \right\}.$$

Every ω with $\Delta(\mu, \omega) > 0$ belongs to $\overline{B}_{R(\mu)}$. If $\Delta(\mu) > 0$, then $B_p > 0$ and $|P_{p,\mu}|$ attains its maximum at some $\omega^* \in \overline{B}_{R(\mu)}$, with $\Delta(\mu, \omega^*) = \Delta(\mu)$. For every ω with $\Delta(\mu, \omega) > 0$, the insertion

$$(52) \quad \kappa(\omega) := -\frac{\Delta(\mu, \omega)}{B_p^2} \operatorname{sign}(P_{p,\mu}(\omega)), \quad c(\omega) := \frac{\kappa(\omega)}{w_p(\omega)},$$

satisfies

$$(53) \quad J(\mu + c(\omega)\delta_\omega) \leq J(\mu) - \frac{\Delta(\mu, \omega)^2}{2B_p^2}.$$

Proof. We first establish attainment of the certificate maximum $\|P_{p,\mu}\|_\infty$ when $\Delta(\mu) > 0$. $p > s_1$ and (4) give $K_p \in C_0(\Omega; H^1(D))$, hence $B_p < \infty$ and $P_{p,\mu} \in C_0(\Omega)$. If $\Delta(\mu) > 0$, then $\|P_{p,\mu}\|_\infty > \alpha L_\phi > 0$. Choose a closed ball outside which $|P_{p,\mu}| < \|P_{p,\mu}\|_\infty/2$; this is possible because $P_{p,\mu}$ vanishes at infinity. The continuous function $|P_{p,\mu}|$ attains its maximum on that compact ball. Moreover, $B_p = 0$ would imply $P_{p,\mu} = 0$, so $B_p > 0$.

We now bound the change caused by an arbitrary normalized insertion $\kappa\delta_\omega$ with $\kappa \in \mathbb{R}$. The explicit atomic–continuous representation of Φ_ϕ , the monotonicity and subadditivity of ϕ , and Lemma 3.6 give

$$(54) \quad \begin{aligned} \Phi_\phi(\mu_p + \kappa\delta_\omega) - \Phi_\phi(\mu_p) &\leq \phi(|\kappa|) && \text{by subadditivity at the point } \omega \\ &\leq L_\phi|\kappa| && \text{by the tangent bound at zero.} \end{aligned}$$

Since $(\mu + \kappa w_p(\omega)^{-1}\delta_\omega)_p = \mu_p + \kappa\delta_\omega$, the quadratic fidelity has the exact expansion

$$(55) \quad \begin{aligned} J\left(\mu + \frac{\kappa}{w_p(\omega)}\delta_\omega\right) - J(\mu) &\leq \kappa P_{p,\mu}(\omega) + \frac{\kappa^2}{2}\|K_p(\omega)\|_{H^1(D)}^2 + \alpha L_\phi|\kappa| && \text{by (54)} \\ &\leq \kappa P_{p,\mu}(\omega) + \frac{B_p^2\kappa^2}{2} + \alpha L_\phi|\kappa| && \text{by the definition of } B_p. \end{aligned}$$

Fix $\omega \in \Omega$ with $\Delta(\mu, \omega) > 0$. Then $|P_{p,\mu}(\omega)| = \alpha L_\phi + \Delta(\mu, \omega)$, and substitution of $\kappa = \kappa(\omega)$ from (52) into (55) yields

$$\begin{aligned} J(\mu + c(\omega)\delta_\omega) - J(\mu) &\leq -|\kappa(\omega)|(\alpha L_\phi + \Delta(\mu, \omega)) + \frac{B_p^2|\kappa(\omega)|^2}{2} + \alpha L_\phi|\kappa(\omega)| && \text{since } \operatorname{sign} \kappa(\omega) = -\operatorname{sign} P_{p,\mu}(\omega) \\ &= -|\kappa(\omega)|\Delta(\mu, \omega) + \frac{B_p^2|\kappa(\omega)|^2}{2} && \text{the threshold terms cancel} \\ &= -\frac{\Delta(\mu, \omega)^2}{2B_p^2} && \text{by the choice of } \kappa(\omega). \end{aligned}$$

This proves (53); at $\omega = \omega^*$ it gives the decrease $\Delta(\mu)^2/(2B_p^2)$.

Finally, we derive the radius from (4). If $|\omega| \geq 1$, then Cauchy–Schwarz gives

$$\begin{aligned} |P_{p,\mu}(\omega)| &\leq \|r_\mu\|_{H^1(D)}\|K_p(\omega)\|_{H^1(D)} && \text{by Cauchy–Schwarz} \\ &\leq C_K\|r_\mu\|_{H^1(D)}\frac{(1+|\omega|)^{s_1}}{1+|\omega|^p} && \text{by (4)} \\ &\leq 2^{s_1}C_K\|r_\mu\|_{H^1(D)}|\omega|^{-(p-s_1)} && \text{since } |\omega| \geq 1. \end{aligned}$$

If $|\omega| > R(\mu)$, the last expression is strictly smaller than $\alpha L_\phi/2$ by (51). This proves the claimed localization of every accepted candidate and every maximizer. $\square$

Corollary 5.2. *Let $(\mu_k)_{k \geq 0}$ be finite atomic iterates with $J(\mu_0) < \infty$, and write $\Delta_k := \Delta(\mu_k)$. If $\Delta_k > 0$, insert at an exact maximizer given by Theorem 5.1. Then correct the coefficients, accepting the corrected iterate only if it does not increase the objective. Then*

$$\begin{aligned} \sum_{k=0}^{\infty} \Delta_k^2 &\leq 2B_p^2 J(\mu_0), \\ \Delta_k &\longrightarrow 0, \\ \min_{0 \leq k < n} \Delta_k &\leq B_p \sqrt{\frac{2J(\mu_0)}{n}} \quad (n \geq 1). \end{aligned}$$

These conclusions do not assert convergence of (μ_k) to a global minimizer or an objective-gap rate.

Proof. If $B_p = 0$, then $\Delta_k = 0$ for every k and the claims are immediate. Otherwise Theorem 5.1 and the correction hypothesis give, for every $n \geq 1$,

$$\sum_{k=0}^{n-1} \frac{\Delta_k^2}{2B_p^2} \leq J(\mu_0) - J(\mu_n) \leq J(\mu_0).$$

Letting $n \rightarrow \infty$ gives the first claim, and summability gives $\Delta_k \rightarrow 0$. Finally,

$$\min_{0 \leq k < n} \Delta_k^2 \leq \frac{1}{n} \sum_{k=0}^{n-1} \Delta_k^2 \leq \frac{2B_p^2 J(\mu_0)}{n},$$

which proves the stated rate. $\square$

For the sampled data, Algorithm 1 minimizes

$$(56) \quad J^M(\boldsymbol{\omega}, c) := l^M(\mu_{\boldsymbol{\omega}, c}) + \alpha \sum_{n=1}^N \phi_{\gamma}(w_p(\omega_n) | c_n|).$$

Its normalized insertion profile is

$$P_{p, \mu_t}^M(\omega) := \frac{P_{\mu_t}^M(\omega)}{w_p(\omega)}.$$

For fixed nodes, introduce the normalized coefficients $u_n := w_p(\omega_n) c_n$. The outer-weight correction then minimizes

$$l^M \left(\sum_{n=1}^N \frac{u_n}{w_p(\omega_n)} \delta_{\omega_n} \right) + \alpha \sum_{n=1}^N \phi_{\gamma}(|u_n|).$$

Theorem 5.1 and Corollary 5.2 apply verbatim with the sampled value–gradient inner product in place of the $H^1(D)$ inner product. The empirical search radius is therefore given by (51), using the sampled residual norm and the corresponding sampled feature-growth constants.

The insertion step. The theorem requires finding a maximizer of $|P_{p, \mu_t}^M(\omega)|$. Finding a global maximizer is generally intractable; for ReLU dictionaries it is NP-hard [2, Section 3.3]. The practical multistart search is proposed in the PDAP algorithm in the paper [20]. At the current iterate $\mu_t = \mu_{\boldsymbol{\omega}^{(t)}, c^{(t)}}$ a candidate ω is therefore accepted when

$$(57) \quad \Delta(\mu_t, \omega) > 0, \quad \text{that is,} \quad |P_{p, \mu_t}^M(\omega)| > \alpha L_{\phi}.$$

Theorem 5.1 provides two additional pieces of information. First, it gives the bound $R(\mu_t)$ for sampling the initial points at each iteration. Second, it gives an initial outer weight that guarantees a single-atom insertion decrease before the local semismooth Newton correction.

For a nonhomogeneous activation set $R_{\text{search}}(\mu_t) := \min\{R(\mu_t), e^5\}$. Each multistart initial point has norm at most $R_{\text{search}}(\mu_t)$. Starting from each ω_0 , one unconstrained L-BFGS solve jointly optimizes all components of ω for $|P_{p, \mu_t}^M(\omega)|$. After optimization, candidates with $|\omega| > R_{\text{search}}(\mu_t)$ are discarded. This avoids constrained optimization.

Before running the SSN solver, the algorithm calculates

$$(58) \quad A_p^M(\omega) := \left\| \frac{K(\omega)}{w_p(\omega)} \right\|_M^2, \quad \kappa_A(\omega) := -\frac{\Delta(\mu_t, \omega)}{A_p^M(\omega)} \text{sign}(P_{p, \mu_t}^M(\omega)), \quad c_A(\omega) := \frac{\kappa_A(\omega)}{w_p(\omega)}.$$

The coefficient $c_A(\omega)$ is used to initialize the subsequent semismooth Newton correction, which optimizes all outer weights. The decrease statement holds for every accepted candidate, not only for a global maximizer of $|P_{p, \mu_t}^M|$.

Writing $h_\gamma(z) := \phi_\gamma(|z|) - |z|$ and $\tau = (1 + \alpha\gamma)^{-1}$, the semismooth Newton solve uses the coordinatewise residual

$$(59) \quad u = \text{prox}_{\tau\alpha|\cdot|} \left(u - \tau \left[\nabla_{ul}^M \left(\sum_{n=1}^N \frac{u_n}{w_p(\omega_n)} \delta_{\omega_n} \right) + \alpha \nabla h_\gamma(u) \right] \right).$$

After the solve, the physical outer weights are recovered as $c_n = u_n/w_p(\omega_n)$; the proximal map in (59) is ordinary soft thresholding.

Algorithm 1 uses the scalar penalty ϕ_γ from (32) and the objective (56), with the insertion condition (57).

Algorithm 1 Iterative node insertion for J^M

- 1: Choose T_{out} , N_{trial} , N_{ins} , $\varepsilon_{\text{prune}}$, and an initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Set $R_{\text{search}} = \min\{R(\mu_t), e^5\}$. Sample N_{trial} points ω_0 with norm bounded by R_{search} .
 - 4: From each ω_0 , run one unconstrained joint L-BFGS maximization of $|P_{p, \mu_t}^M(\omega)|$.
 - 5: Discard candidates with $|\omega| > R_{\text{search}}$, then remove near-duplicates.
 - 6: Keep the candidates satisfying (57); terminate if there is none.
 - 7: Insert at most N_{ins} nodes with the largest $\Delta(\mu_t, \omega)$, warm-start outer weight by (58).
 - 8: Optimize the outer weights of the enlarged network by the semismooth Newton method.
If the correction increases J^M , discard it and keep the post-insertion weights.
 - 9: Remove nodes with $|c_n| \leq \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
 - 10: $t = t + 1$.
 - 11: **end while**
-

5.2. Algorithm 2: the fractional-exponent penalty on the sphere. Algorithm 2 uses $\psi_k(z) = z^q$, $q = 2/(k+1)$, on the unit sphere. For $q < 1$, insertion condition of Algorithm 1 is unavailable (Remark 4.8). Furthermore, since the zero network is a local minimizer and the infinitesimal, we need good initialization method, which is provided by the Theorem 4.7

We therefore test the actual change produced by inserting one atom with a finite coefficient. We use the scalar proximal map

$$(60) \quad \text{prox}_{\lambda|\cdot|^q}(v) \in \arg \min_{z \in \mathbb{R}} \left\{ \frac{1}{2}(z-v)^2 + \lambda|z|^q \right\},$$

with zero selected when the minimum is attained both at zero and at a nonzero point. For $0 < q < 1$, put

$$\bar{z} = [2\lambda(1-q)]^{1/(2-q)}, \quad \bar{v} = \frac{2-q}{2(1-q)} \bar{z}.$$

Then the selected proximal value is zero when $|v| \leq \bar{v}$; when $|v| > \bar{v}$, it is $\text{sign}(v)z$, where z is the largest positive solution of

$$z + \lambda q z^{q-1} = |v|.$$

The implementation evaluates this value in closed form for $q \in \{1/2, 2/3\}$, using respectively a cubic in $\sqrt{z}$ and a quartic in $z^{1/3}$. At the endpoint $q = 1$, (60) is ordinary soft thresholding.

For a candidate $\omega \notin \text{atom } \mu_t$, write

$$P := P_{\mu_t}^M(\omega), \quad A := \|K(\omega)\|_M^2.$$

If A or $|P|$ is numerically zero, the candidate is discarded. Otherwise the exact one-atom objective increment is

$$(61) \quad \Delta J_{\psi_k}^M(c; \omega) = cP + \frac{1}{2}Ac^2 + \alpha|c|^q.$$

Completing the square in (61) gives its selected global minimizer directly:

$$(62) \quad c^*(\omega) := \text{prox}_{(\alpha/A)|\cdot|^q} \left(-\frac{P}{A} \right).$$

We accept the candidate precisely when $c^*(\omega) \neq 0$ and $\Delta J_{\psi_k}^M(c^*(\omega); \omega) < 0$. Thus the same coefficient both certifies the finite-step decrease and initializes the subsequent correction.

After insertion, let m_t be the smallest magnitude among the nonzero outer weights. For $q < 1$, the correction fixes the proximal scale

$$(63) \quad \lambda_t := \min \left\{ \alpha, \frac{\rho_{\text{prox}} m_t^{2-q}}{2(1-q)} \right\}, \quad \rho_{\text{prox}} = 0.5,$$

for the whole semismooth Newton solve. Every nonzero warm-start coefficient is strictly above the switching magnitude, since $\bar{z} \leq \rho_{\text{prox}}^{1/(2-q)} m_t < m_t$. Define, coordinatewise,

$$(64) \quad c(z) := \text{prox}_{\lambda_t|\cdot|^q}(z), \quad G_t(z) := \frac{\alpha}{\lambda_t}(z - c(z)) + \nabla_c l^M(\mu_{\omega, c(z)}).$$

The coefficient correction applies semismooth Newton to $G_t(z) = 0$. For $q = 1$ we use soft thresholding with $\lambda_t = \alpha$. For $q < 1$, an exact undamped Newton iteration on a fixed smooth branch has superlinear convergence rate locally under nonsingularity.

The implementation uses $k \in \{2, 3\}$, corresponding to $q \in \{2/3, 1/2\}$. The ReLU- ℓ^1 baseline reported alongside it is not this algorithm at $k = 1$: it is the classical profile-threshold insertion with a coordinate-descent warm start and the convex penalty $\alpha \sum_i |c_i|$, run separately.

Algorithm 2 Iterative node insertion for $J_{\psi_k}^M$

- 1: Choose T_{out} , N_{trial} , N_{ins} , $\varepsilon_{\text{prune}}$, and an initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Sample N_{trial} nodes randomly as the initial points.
 - 4: For each initial point, run unconstrained L-BFGS w.r.t. $|P|$ and normalize onto $\mathbb{S}^d$.
 - 5: Remove cosine near-duplicates and discard final candidates that coincide numerically with the current support.
 - 6: For each candidate compute $c^*(\omega)$ from (62), and keep it when $c^*(\omega) \neq 0$ and $\Delta J_{\psi_k}^M(c^*(\omega); \omega) < 0$; terminate if there is none.
 - 7: Retain at most N_{ins} candidates with the smallest $\Delta J_{\psi_k}^M(c^*; \omega)$, insert with outer weight c^* .
 - 8: Apply the semismooth Newton correction (64), using (63). If the correction increases $J_{\psi_k}^M$, discard it and keep the inserted coefficients c^* .
 - 9: Remove nodes with $|c_n| \leq \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
 - 10: $t = t + 1$.
 - 11: **end while**
-

6. NUMERICAL EXAMPLES

The experiments use seed 42. Each reported parameter choice is run once, and the validation sets are random subsets of the stored sample rows, with 90% used for training and 10% for validation. Input and value normalization are computed from the full sample before the split.

For a fitted value function $\widehat{V}$, the relative errors are

$$\begin{aligned} \text{Err}_{L^2} &:= \frac{\|\widehat{V} - V\|_{L^2}}{\|V\|_{L^2}}, \\ \text{Err}_{H^1} &:= \frac{(\|\widehat{V} - V\|_{L^2}^2 + \|\nabla \widehat{V} - \nabla V\|_{L^2}^2)^{1/2}}{(\|V\|_{L^2}^2 + \|\nabla V\|_{L^2}^2)^{1/2}}. \end{aligned}$$

Both norms are computed empirically on the validation or evaluation set specified for each experiment with the original scale before normalization. For a frontier, support sizes are sorted and the reported error is the smallest among recorded iterates with no more atoms; it is not an iteration trace.

6.1. Van der Pol Oscillator. We consider the optimal control of the Van der Pol oscillator:

$$\min_{u \in L^2(0, T; \mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^T (y_1^2(t) + y_2^2(t) + \eta u^2(t)) dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1 - y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

In this setting, $d = 2$, $m_u = 1$, $\eta = 0.1$, $T = 3$, and

$$f(y) = \begin{pmatrix} y_2 \\ -y_1 + y_2(1 - y_1^2) \end{pmatrix}, \quad g(y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad h(y) = y_1^2 + y_2^2.$$

By the optimality condition, $u^* = -\frac{1}{2\eta} g(y^*)^\top p^* = -\frac{1}{2\eta} p_2^*$. The two-point boundary value problem (TPBVP) becomes

$$(65) \quad \begin{cases} \dot{y}_1^* = y_2^*, \\ \dot{y}_2^* = -y_1^* + y_2^*(1 - y_1^{*2}) + u^*, \\ -\dot{p}_1^* = 2y_1^* - p_2^*(1 + 2y_1^*y_2^*), \\ -\dot{p}_2^* = 2y_2^* + p_1^* + p_2^*(1 - y_1^{*2}), \\ y^*(0) = x, \quad p^*(T) = 0. \end{cases}$$

We solve (65) by adaptive collocation at each point of a 30×30 grid on $[-3, 3]^2$, using a residual tolerance of 10^{-8} . At each state we retain the lower-cost converged solution from two initial guesses. Three-point Gauss quadrature on each adaptive mesh interval gives the value label, using the full running cost stated above; the initial costate gives the gradient label $g = p(0)$. These are numerical PMP references, without a certificate of global optimality. All 900 solves converge. The resulting dataset is split into 810 training and 90 validation rows.

The Algorithm 1, Algorithm 2, and moment-order fits use at most $T_{\text{out}} = 150$ outer iterations and $N_{\text{trial}} = 50$ sampled candidate nodes per iteration. Each iteration inserts the single selected candidate ($N_{\text{ins}} = 1$), so the one-candidate decrease statement applies. Corollary 5.2 remains conditional on the selected candidate being an exact global maximizer and is not claimed for the multistart search. The loop terminates when no candidate clears the insertion threshold. The separate traditional ReLU- ℓ^1 comparators use 10 outer iterations with batch insertion of at most 15 candidates per iteration, from 50 candidate starts. Their insertion schedule therefore differs from the sequential nonconvex fits.

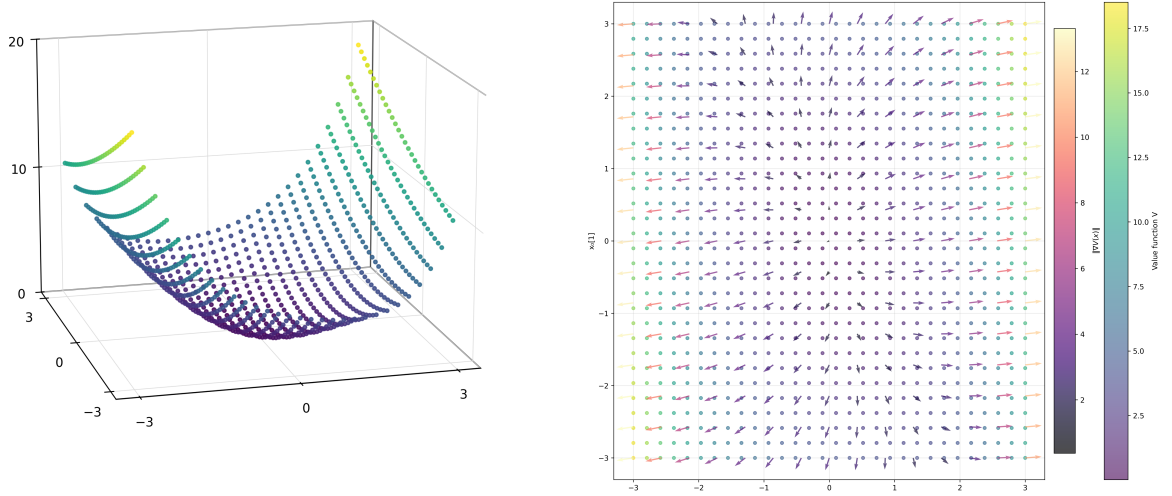


FIGURE 2. Van der Pol numerical value and gradient samples over $[-3, 3]^2$. Left: value labels v^m . Right: v^m (dot color) and costate labels g^m (arrows); arrow color indicates $|g^m|$.

6.1.1. *Effect of the moment order p .* The moment order enters the objective through the weight $w_p(\omega) = 1 + |\omega|^p$ inside the scalar penalty: an atom at ω is charged $\phi_\gamma(w_p(\omega)|c|)$ rather than $\phi_\gamma(|c|)$, so a fixed coefficient becomes more expensive the further the atom sits from the origin. The exponent p is therefore an active parameter of the objective, not a device that acts only through a separate coefficient.

Observed support radius. For a finite atomic μ write

$$R_{\max}(\mu) := \max\{|\omega| : \omega \in \text{atom } \mu\}$$

for the radius of its support. Lemma 3.19 localizes the support of local minimizers under its assumptions, motivating this measurement. The finite empirical fits below are not certified local minimizers of that problem; their reported radii are measured directly. We use Algorithm 1 with $\gamma = 1$, equal value and gradient weights, and seed 42, for softplus, tanh, the Gaussian and GELU², over

$$\alpha \in \{10^{-4}, 10^{-5}\}, \quad p \in \{2.01, 2.5, 3, 4\},$$

on both benchmarks. Figure 3 shows the resulting sixteen (problem, activation, α) families. In fifteen families $R_{\max}$ decreases monotonically over the tested orders. The largest absolute contraction is for GELU² on the pendulum at $\alpha = 10^{-5}$, from 148.06 at $p = 2.01$ to 7.87 at $p = 4$; across all families the largest radius at $p = 4$ is 14.80.

We also record the radius $R_{0.95}$, defined for a discrete measure $\mu = \sum_j c_j \delta_{\omega_j}$ by

$$R_{0.95}(\mu) := \inf \{R \geq 0 : |\mu|(B_R) \geq 0.95 |\mu|(\Omega)\},$$

and the support size N . Unlike $R_{\max}$, both depend on the fitted coefficient distribution. The radius $R_{0.95}$ decreases monotonically in thirteen of the sixteen families, and the support size in twelve, so their behavior is reported separately rather than inferred from the support theorem.

The computable search radius. Theorem 5.1 supplies the radius (51) outside which no candidate can satisfy the insertion condition. Its constants are available before the fit: the growth exponents s_0, s_1 and the constant C_ρ of Assumption 3.1 are properties of the activation, and the remaining factor follows from the sample extent, so $C = C_\rho(A_M^{s_0} + A_M^{s_1-1})$ with $A_M = \max\{1, \max_m \sqrt{1 + |x^m|^2}\}$. Nothing is calibrated from the fit.

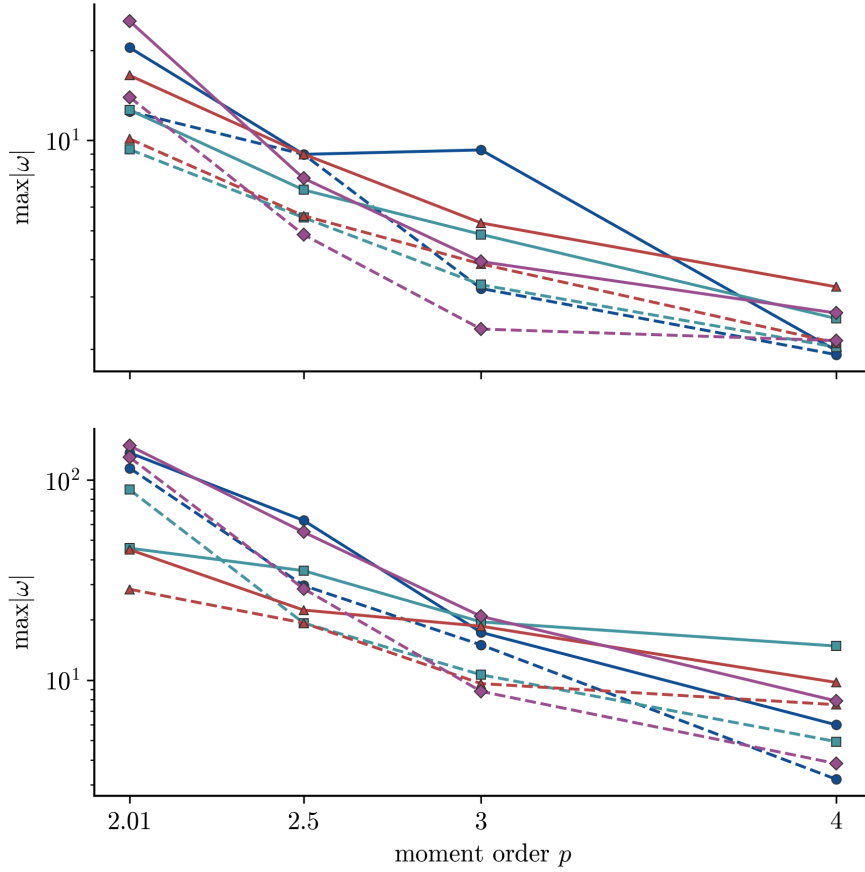


FIGURE 3. Support radius $R_{\max}$ against the moment order p , for softplus, tanh, the Gaussian and GELU² at $\alpha = 10^{-4}$ and $\alpha = 10^{-5}$; upper panel Van der Pol, lower panel pendulum swing-up. Blue circles denote softplus, teal squares tanh, red triangles the Gaussian, and purple diamonds GELU². Solid lines use $\alpha = 10^{-5}$ and dashed lines $\alpha = 10^{-4}$. All curves use seed 42; fifteen of the sixteen decrease monotonically.

On the normalized Van der Pol training inputs ($A_M = \sqrt{3}$, hence $C = 2.73$ for softplus), the search radius at initialization is 48.09 at $p = 4$, $\alpha = 10^{-4}$ and 103.61 at $p = 4$, $\alpha = 10^{-5}$. It decreases during these fits to 22.05 and 43.89, respectively. The implementation also imposes the fixed bound $\lambda \leq e^5$, so the theorem radius changes the search only when it is smaller than that bound.

The recorded search-radius histories fall below e^5 in thirteen of the thirty-two Van der Pol moment-order runs: one at $p = 2.5$, five at $p = 3$, and seven at $p = 4$. The $p = 2.5$ case is the Gaussian at $\alpha = 10^{-4}$. At $p = 3$, both tanh and the Gaussian activate the smaller cap at both values of α , as does softplus at $\alpha = 10^{-4}$. At $p = 4$, all three activate it at both values of α , together with GELU² at $\alpha = 10^{-4}$. These runs do not include a paired fixed-cap comparison, so they do not establish an improvement or degradation in fit caused by applying the theorem radius.

6.1.2. Algorithm 1: the log penalty on the normalized measure. We investigate the effect of including gradient data for softplus, tanh, and the Gaussian. Table 1 compares fitting the values alone with fitting both values and gradients.

TABLE 1. Effect of gradient-augmented fitting with $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$, seed 42. The same parameters are used for all three activations, so the comparison is not confounded with per-activation tuning.

Activation	L^2 trained	H^1 trained	Activation	L^2 trained	H^1 trained
softplus	8.38×10^{-2}	1.24×10^{-2}	softplus	2.23×10^{-1}	3.11×10^{-2}
tanh	7.59×10^{-2}	5.85×10^{-3}	tanh	2.41×10^{-1}	2.18×10^{-2}
Gaussian	5.28×10^{-2}	2.09×10^{-3}	Gaussian	1.82×10^{-1}	1.11×10^{-2}

(A) Effect of gradient training on Err_{L^2} (B) Effect of gradient training on Err_{H^1}

Choice of activation function in gradient training. We compare softplus, tanh, and the Gaussian. Because the gradient-augmented loss contains both the value and gradient errors, $K(\omega)$ contributes to the approximation of V and $\nabla K(\omega)$ contributes to the approximation of ∇V . Figure 4 contrasts the three activations through their value, first derivative, and second derivative. The softplus derivative increases from 0 to 1, the derivative of tanh tends to zero at both ends of the real line, and the Gaussian derivative is localized and changes sign. These derivatives generate different columns $a\rho'(a \cdot x + b)$ in the gradient fitting problem.

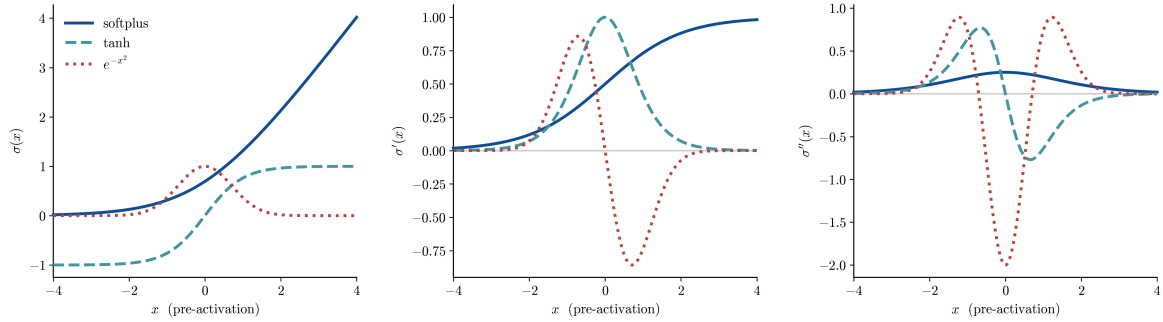


FIGURE 4. The three smooth profiles used by Algorithm 1 — softplus, tanh, and the Gaussian e^{-x^2} — shown as ρ , ρ' , and ρ'' . Under the gradient-augmented loss, the first derivative is the basis that reconstructs ∇V .

For the tested parameters, the Gaussian has the smallest relative H^1 validation error, while softplus uses the fewest atoms. Thus accuracy and support size give different rankings.

TABLE 2. Support size and validation error for the parameter values in Table 1.

Activation	N	Err_{L^2}	Activation	N	Err_{H^1}
softplus	6	8.38×10^{-2}	softplus	14	3.11×10^{-2}
tanh	10	7.59×10^{-2}	tanh	29	2.18×10^{-2}
Gaussian	10	5.28×10^{-2}	Gaussian	32	1.11×10^{-2}

(A) L^2 trained (B) H^1 trained

With the H^1 loss, all three relative gradient errors fall substantially: from 0.328 to 0.0453 for softplus, from 0.358 to 0.0327 for tanh, and from 0.273 to 0.0169 for the Gaussian. Value

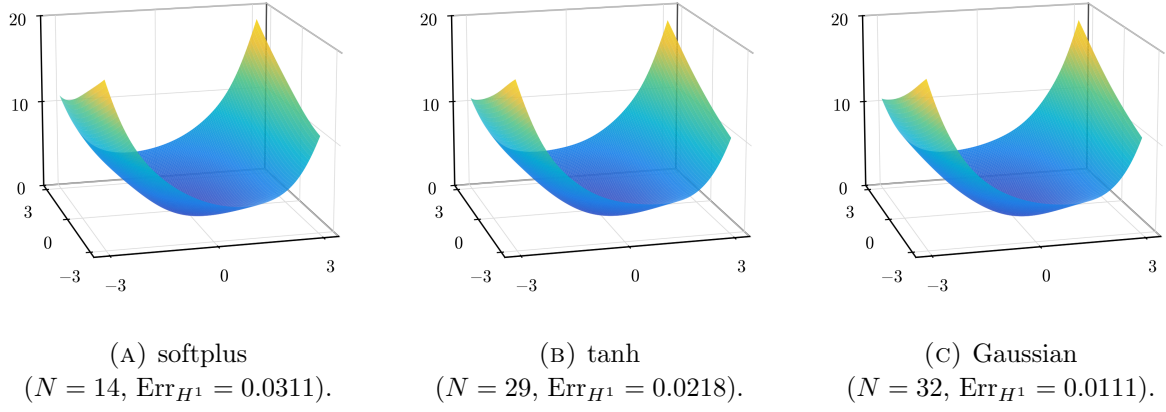


FIGURE 5. Learned value functions under the H^1 loss using the parameter values in Table 1. All three fit the numerical reference data; they differ in support size and validation error. Predictions are clipped to $[0, 20]$ for visualization; the reported errors use the unclipped network outputs.

accuracy improves as well (Table 1). The Gaussian reaches relative H^1 error 0.0111 with 32 atoms; tanh reaches 0.0218 with 29, and softplus 0.0311 with 14.

6.1.3. *Algorithm 2: $|c|^q$ -penalty with ReLU^k activation.* Algorithm 2 uses the profile $\rho(z) = \text{ReLU}(z)^k$ with the penalty ψ_k , namely $\alpha \sum_i |c_i|^q$ with $q = 2/(k+1)$. The swept parameter is the power k ; all fits below use $\alpha = 10^{-5}$.

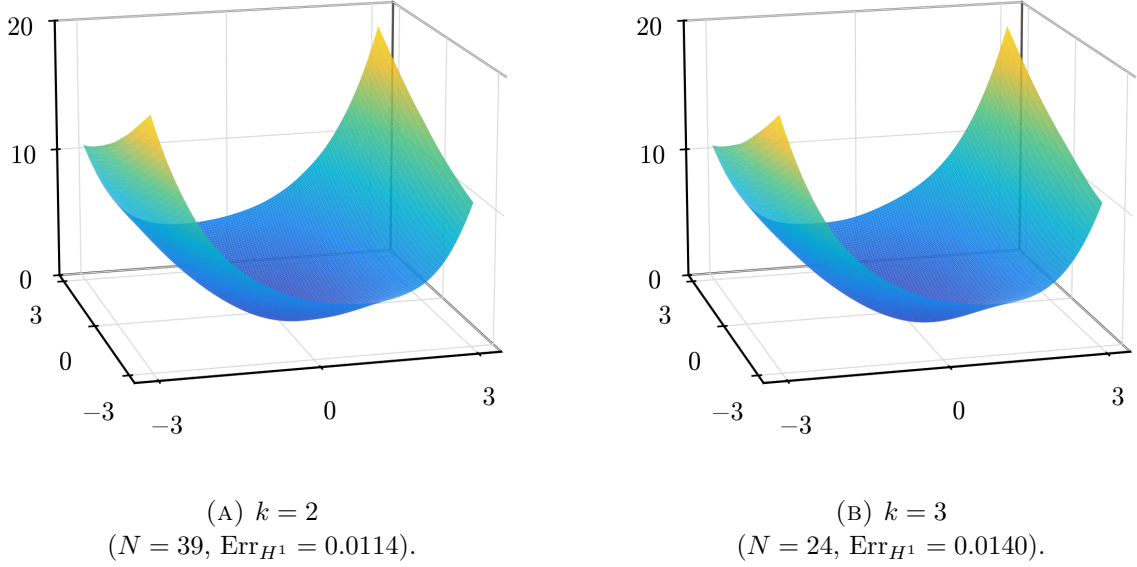


FIGURE 6. H^1 -trained networks with $\alpha = 10^{-5}$ for $k \in \{2, 3\}$. Predictions are clipped to $[0, 20]$ for visualization; the reported errors use the unclipped network outputs.

For these fixed- α , single-seed fits, adding gradient data lowers the combined value–gradient error from 0.0813 to 0.0114 for $k = 2$, and from 0.0875 to 0.0140 for $k = 3$, factors of about 7.13 and 6.26, respectively (Table 3). Value error also decreases, while the support becomes larger. This comparison alone does not establish better closed-loop performance.

TABLE 3. Algorithm 2: network size and validation error vs. the atom power k ($\alpha = 10^{-5}$).

Power	N	Err_{L^2}	Err_{H^1}		Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	14	2.55×10^{-2}	8.13×10^{-2}	(A) L^2 trained	$k = 2$	39	2.75×10^{-3}	1.14×10^{-2}
$k = 3$	13	2.79×10^{-2}	8.75×10^{-2}		$k = 3$	24	5.24×10^{-3}	1.40×10^{-2}
				(B) H^1 trained				

Changing k changes both the activation and the induced exponent $q = 2/(k+1)$. At comparable H^1 error, the $k = 3$ fit uses 24 atoms instead of 39 for $k = 2$. Under the L^2 loss the fits use 14 and 13 atoms, respectively, with relative H^1 errors 0.0813 and 0.0875 (Table 3(a)).

6.1.4. *Summary.* Figure 7 plots the best observed relative H^1 error at each support budget along the selected fits. At the shared Algorithm 1 operating point, softplus uses 14 atoms at error 0.0311 and the Gaussian uses 32 at 0.0111. At $\alpha = 10^{-6}$, the selected minimum-training-objective Algorithm 2 checkpoints use 70 atoms at 0.00559 for ReLU^2 and 30 atoms at 0.00860 for ReLU^3 .

For a separate long-horizon stabilization diagnostic, we synthesize feedback laws from $y_0 = (2, 1)$ and roll them out over $T = 12$ with fixed-step RK4, zero-order-hold controls clipped to $|u| \leq 50$ as

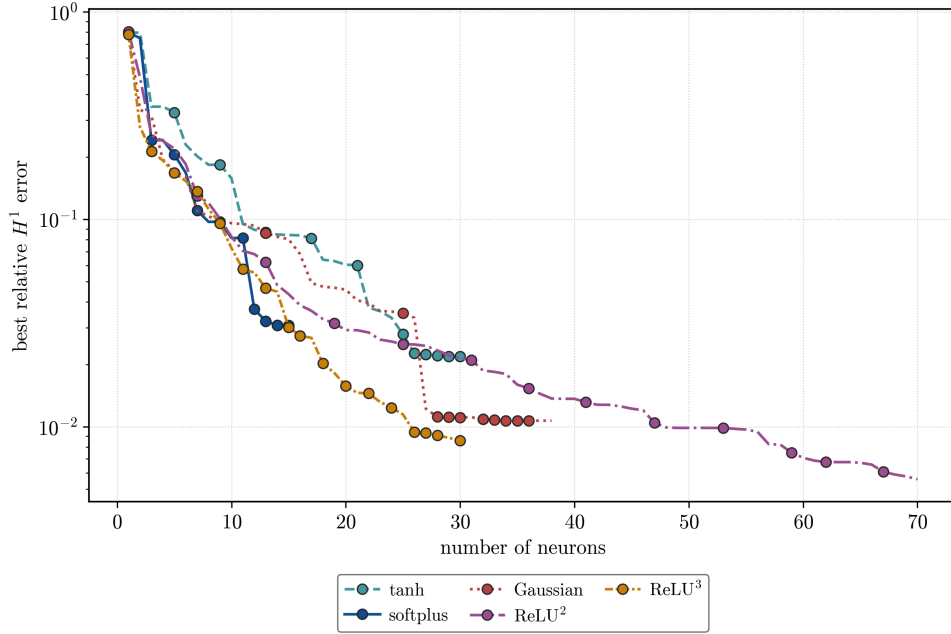


FIGURE 7. Best observed relative H^1 validation error against support budget. Nonhomogeneous profiles use the parameter values stated in Table 1; homogeneous profiles use $\alpha\psi_k$ at $\alpha = 10^{-6}$ for both $k = 2$ and $k = 3$.

a numerical guard. The reference law interpolates the time-zero costate component of the dataset with a C^1 Clough–Tocher interpolant and applies it as a stationary feedback. The finite-horizon optimal feedback is time-dependent and the rollout horizon exceeds the data horizon $T = 3$. Therefore its rollout cost 6.46 is not $V(0, y_0)$, and the comparison is not a test of finite-horizon optimality. Figure 8 shows $\|y(t)\|$ and $|u(t)|$ for softplus, the Gaussian, and ReLU^k ($k = 3$), beside this reference. All three reach $\|y(T)\| < 0.2$, with RK4-stage rollout costs 6.46, 6.46, and 6.48, respectively, compared with 6.46 for the interpolated reference. This terminal-neighbourhood criterion does not imply convergence to the origin.

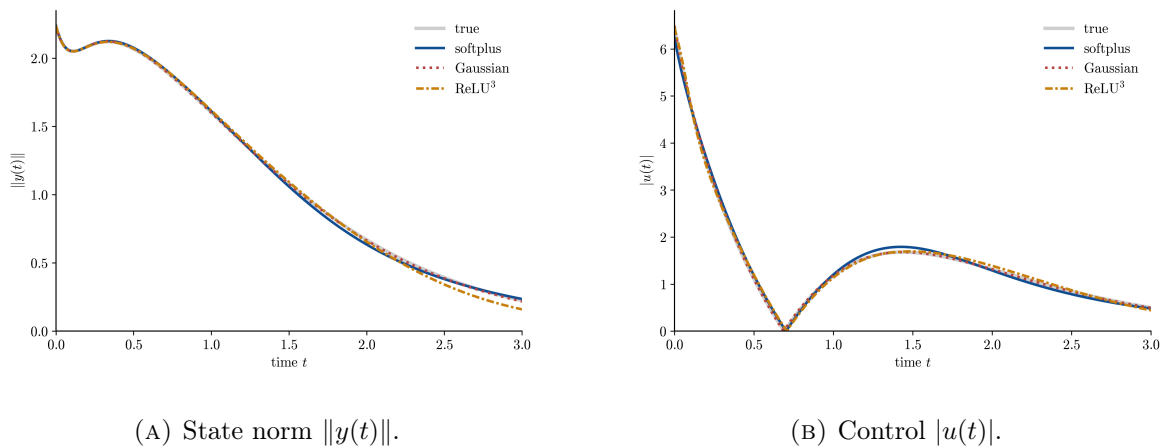


FIGURE 8. First three seconds of the $T = 12$ stabilization diagnostic from $y_0 = (2, 1)$ under the feedback synthesized from each fitted value function and under the stationary reference law obtained from the time-zero costates.

We additionally compare all five fitted controllers with a numerical open-loop solve from the same initial state over the original benchmark horizon $T = 3$ (Figure 9). Each fitted law

is evaluated along its own simulated trajectory. The reference solves the finite-horizon PMP boundary-value problem with free terminal state and zero terminal cost, so that $p(T) = 0$. We initialize the solve from the learned and dataset-feedback rollouts and retain the least-cost converged solution; this procedure does not certify global optimality.

The compared networks have 29 neurons for tanh, 14 for softplus, 32 for the Gaussian, 70 for ReLU^2 , and 30 for ReLU^3 . The nonhomogeneous fits use $(\alpha, \gamma, p) = (10^{-4}, 10, 2.01)$, and both homogeneous fits use $\alpha = 10^{-6}$. The same checkpoints supply the summary widths and feedback diagnostics above.

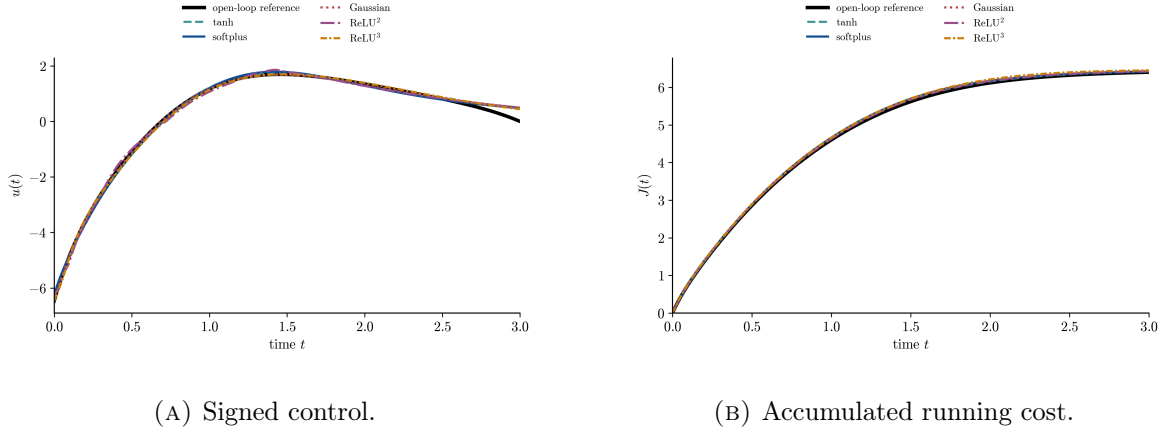


FIGURE 9. Van der Pol controls and realized costs from $y_0 = (2, 1)$ over $T = 3$. The black curve is the numerical open-loop reference. Each colored curve uses its own state-control trajectory, with zero-order-hold controls and the numerical bound $|u| \leq 50$. The accumulated cost $J(t) = \int_0^t \ell(y(s), u(s)) ds$ uses the held-control RK4 stages with $\Delta t = 0.01$ for learned laws and three-point Gauss quadrature on the dense PMP solution for the reference. It has no learned terminal-value contribution. The preceding stabilization diagnostic instead uses $T = 12$ with the same quadrature. Controls at equal times are generally evaluated at different states.

Figure 10 shows the inner parameters for the Gaussian, softplus, and ReLU^k ($k = 3$). The two algorithms produce structurally different representations: Algorithm 2 constrains its atoms to the unit sphere $\mathbb{S}^2$, while Algorithm 1 leaves them free — the Gaussian atoms in particular span a wide range of norms.

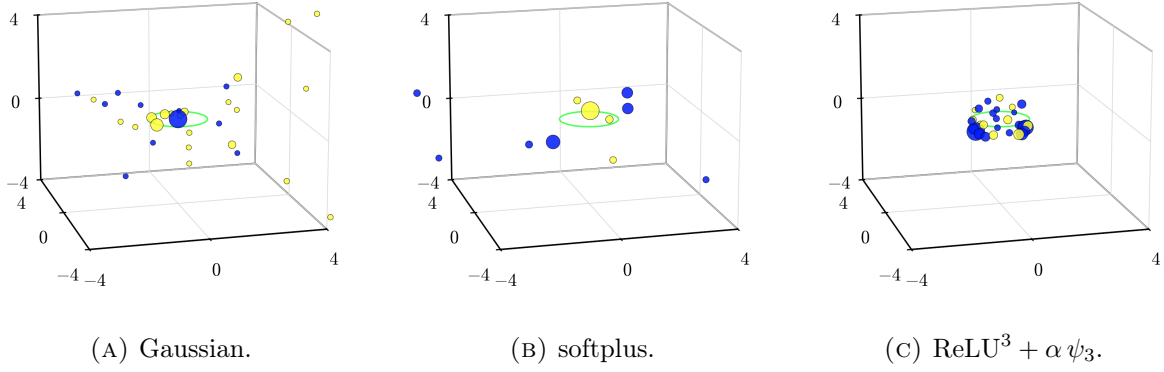


FIGURE 10. Inner parameters (a_1, a_2, b) , with the unit circle in the plane $b = 0$ shown in light green. The two nonhomogeneous fits use $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$; dot color gives the sign and dot size the magnitude of the outer weight.

6.2. Pendulum Swing-Up. The question in this example is how a sparse gradient-trained network approximates a continuous value function whose gradient jumps across a switching set. A finite network with a continuously differentiable activation has a continuous gradient and therefore cannot reproduce the jump exactly.

The example we consider follows from [10]. We write the state as

$$x = (x_1, x_2)^\top = (\theta, \dot{\theta})^\top,$$

the pendulum angle and angular velocity. (The letter ω remains reserved for the network parameters.) We consider the optimal control problem

$$\min_{u \in L^2(0, \infty; \mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^\infty (\sin^2 \theta(t) + (\cos \theta(t) - 1)^2 + \dot{\theta}^2(t) + \eta u^2(t)) dt$$

subject to

$$\begin{cases} m\ell^2 \ddot{\theta} = -b\dot{\theta} + mg\ell \sin \theta + u, \\ (\theta(0), \dot{\theta}(0)) = (x_1, x_2). \end{cases}$$

The target equilibria are $x_{U,j} := (2j\pi, 0)$, $j \in \mathbb{Z}$; for a selected target we write $x_U = x_{U,j}$.

In this setting the state dimension is $d = 2$ and the control is scalar. The physical and cost parameters are $m = \ell = 1$, $b = 0.1$, $g = 9.8$, and $\eta = 1$. To initialize the backward integration, we use the local linear-quadratic regulator (LQR) approximation near x_U . Let $\tilde{x} = x - x_U$ be the local coordinate around the upright equilibrium. Linearizing the dynamics at $(x_U, 0)$ gives

$$\dot{\tilde{x}} = A\tilde{x} + Bu, \quad A = \begin{pmatrix} 0 & 1 \\ \frac{g}{\ell} & -\frac{b}{m\ell^2} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \frac{1}{m\ell^2} \end{pmatrix}.$$

The running cost is locally approximated by

$$|\tilde{x}|^2 + \eta u^2.$$

The local infinite-horizon LQR value function is

$$V^\infty(\tilde{x}) = \tilde{x}^\top P \tilde{x},$$

where P is the positive definite solution of the algebraic Riccati equation

$$A^\top P + PA - \frac{1}{\eta} P B B^\top P + I_2 = 0.$$

The superscript ∞ refers to the infinite-horizon LQR problem. For $\varepsilon > 0$, we define the local LQR sublevel set and boundary by

$$\mathcal{E}_\varepsilon = \{\tilde{x} : V^\infty(\tilde{x}) \leq \varepsilon\}, \quad \partial\mathcal{E}_\varepsilon = \{\tilde{x} : V^\infty(\tilde{x}) = \varepsilon\}.$$

We integrate the nonlinear PMP equations with a numerical LQR terminal initialization:

$$(66) \quad \begin{cases} m\ell^2 \ddot{\theta} = -b\dot{\theta} + mg\ell \sin \theta + u, \\ -\dot{p}_1 = 2 \sin \theta + \frac{g}{\ell} \cos \theta p_2, \\ -\dot{p}_2 = 2\dot{\theta} + p_1 - \frac{b}{m\ell^2} p_2, \\ x(0) = x, \quad x(T_x) - x_U \in \partial\mathcal{E}_\varepsilon, \quad p(T_x) = \nabla V^\infty(x(T_x) - x_U). \end{cases}$$

Here the terminal condition approximates the nonlinear value gradient by the LQR gradient. In the experiment we use $\varepsilon = 2 \times 10^{-4}$ and initialize the backward costate on $\partial\mathcal{E}_\varepsilon$ by

$$p(T_x) = \nabla V^\infty(\tilde{x}(T_x)) = 2P\tilde{x}(T_x).$$

For the data generation, we integrate 2000 PMP paths backward from the local LQR boundary $\partial\mathcal{E}_\varepsilon$. Along each path, the accumulated running cost plus the boundary value ε gives a numerical branch value label v , and the costate gives the gradient label $g = p$. These approximate the nonlinear value and gradient; the LQR matching is not an exact free-arrival terminal condition for the nonlinear problem. The backward integration stops when the accumulated value reaches 100. We use adaptive boundary angles, a maximum integration step of 0.005, and relative and absolute tolerances of 10^{-10} and 10^{-12} . This yields the dataset

$$\{x^m, v^m, g^m\}_{m=1}^M.$$

The construction provides samples on both sides of the gradient discontinuity. For $j \in \mathbb{Z}$, let $V_{2j\pi}(x)$ be the fixed-target infinite-horizon value obtained by minimizing over paths converging to $(2j\pi, 0)$. The stationary value is

$$V(x) = \min_{j \in \mathbb{Z}} V_{2j\pi}(x).$$

In the region where the targets $(0, 0)$ and $(2\pi, 0)$ are minimizing, this reduces to $V(x) = \min\{V_0(x), V_{2\pi}(x)\}$. Define the representative switching set between these adjacent targets by

$$S_{0,2\pi} := \{x \in \mathbb{R}^2 : V_0(x) = V_{2\pi}(x) = V(x)\}.$$

The periodic switching set and its closed δ -neighbourhood are

$$S := \bigcup_{j \in \mathbb{Z}} (S_{0,2\pi} + (2j\pi, 0)), \quad S_\delta := \{x \in \mathbb{R}^2 : \text{dist}(x, S) \leq \delta\}, \quad \delta > 0.$$

The regional errors below use $\delta = 0.3$ and are empirical errors on evaluation samples inside and outside this tube. Numerically, distances are measured to the reconstructed switching points and their $\pm 2\pi$ translations. The square $[-8, 8]^2$ is the visualization window, not a filter on the training or evaluation data; some augmented and evaluation states lie slightly outside it. At points of S , two adjacent fixed-target values agree while their gradients may differ; the sampled PMP costates provide the corresponding one-sided gradients, without asserting a classical gradient on S .

To reconstruct $S_{0,2\pi}$, fix a value level ℓ and interpolate one state with value ℓ on each raw PMP path. Ordering these states by their terminal-boundary angle gives a polygonal characteristic contour. By periodicity, translating it by $(2\pi, 0)$ gives a contour for the adjacent target. Their intersections are numerical candidates for equal branch values. Repeating this construction over the sampled levels and tracking the four intersections nearest $(\pi, 0)$ produces an approximation of the four switching arms; their 2π -translations approximate S . Raw characteristics can include dominated branches, so the contour construction is followed by the branch screen described below. We use value levels spaced by 0.05 up to 70 to reconstruct the basin from the same

2000 paths, and retain each characteristic up to its first exit from this basin. The level 70 is a geometry-construction parameter, not a hard cap on the sampled values.

To remove characteristic segments that have become more expensive than a nearby branch, we compare their values with local first-order estimates from the 32 nearest samples within radius 0.1. Body samples must lie within 0.3 of the lower estimate for their own target and, whenever an adjacent-target estimate is available, at least 0.3 below it. This numerical branch screen retains 881,672 candidate body samples; it does not certify global minimality of the characteristic construction.

After reconstructing S , we augment the dataset with samples from both value branches. By periodicity, translating a raw PMP path by $\pm 2\pi$ in θ produces a path associated with an adjacent upright. Not every translated state belongs to the optimal branch, so we keep only states within distance 0.5 of S for which a local first-order comparison identifies the branch as strictly cheaper than its competitor. The final dataset consists of 3,000 original body samples, 300 near-side padding samples, and 600 translated opposite-side collar samples. Its 3,900 rows supply costate labels from both sides of the gradient discontinuity, with values up to 68.67. Figure 11 shows the resulting dataset.

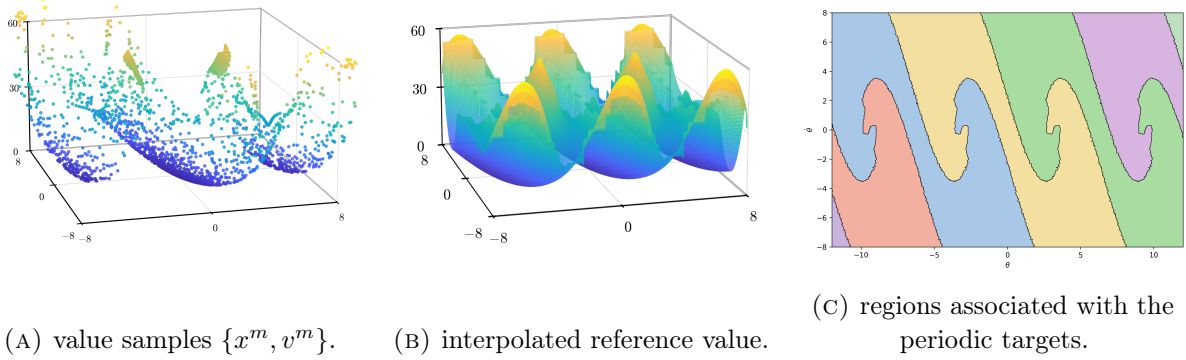


FIGURE 11. Numerical pendulum reference data over the $(\theta, \dot{\theta})$ plane. (a) Selected value samples from the PMP generator. (b) A periodic interpolation of those samples. (c) Target-region visualization obtained by assigning each grid point to the nearest sampled first-exit characteristic and its periodic copies. This coloring approximates the target regions; it does not compare branch costs at every grid point.

6.2.1. Approximation near the switching set. Figure 12 shows the fitted surfaces on the same physical window and value range as the value function in Figure 11b. The cross-section below compares the approximations near S .

Let x_S be a point on the polygonal approximation of S in the most densely sampled neighbourhood, and let n be its estimated unit normal. Figure 13 shows the cross-section $x(s) = x_S + sn$. For each point, we solve two PMP boundary-value problems over $T = 10$, with terminal LQR cost centred at $(0, 0)$ and $(2\pi, 0)$, respectively. Each branch is continued from the end of the section nearer its target. We select the lower computed cost and its initial costate at every point. This yields a numerical approximation of $\min\{V_0, V_{2\pi}\}$; the cost crossing need not coincide exactly with the polygonal location $s = 0$. The smooth profiles have continuous fitted gradients, as do the rectified powers $k \geq 2$, so none can reproduce a classical gradient jump exactly. For $k = 2$ the fitted normal gradient is piecewise linear.

We next ask whether adding samples near S reduces the error on $S_{0.3}$. We compare 6,000-sample sets with augmentation-pool shares of approximately 23%, 40%, and 60%, and a set that

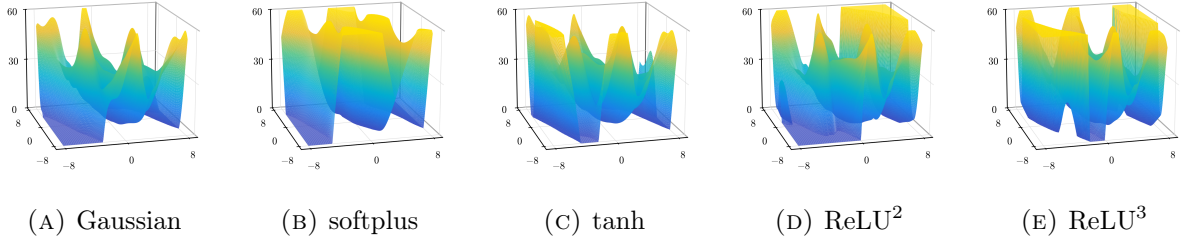


FIGURE 12. Learned value surfaces $\hat{V}(x)$ over the $(\theta, \dot{\theta})$ window $[-8, 8]^2$. Predictions below 0 or above 60 are clipped to the respective boundary for visualization; errors and feedback use the unclipped fitted function. The ReLU^k models are trained with the fractional-exponent penalty ψ_k , and the nonhomogeneous models with (56) using the parameter values reported in Figure 1b.

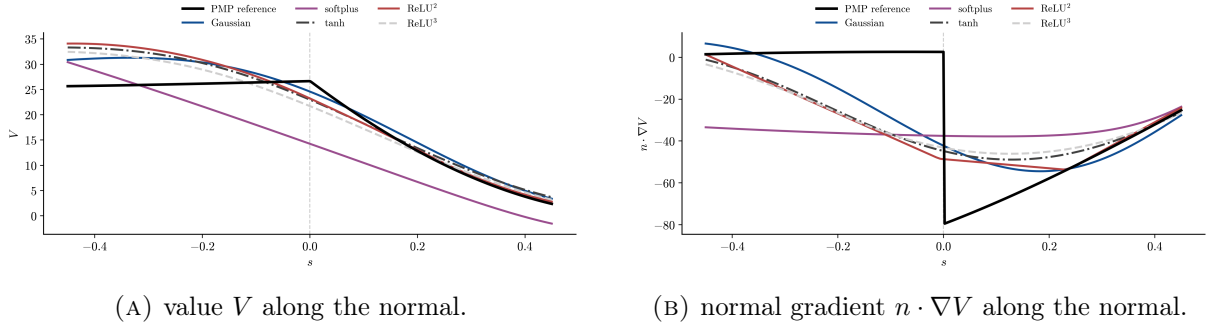


FIGURE 13. Normal cross-section of the reconstructed switching curve, parametrized by signed distance s . The black curve selects the lower-cost of two continued PMP solutions with terminal LQR costs. Left: the reference value and fitted values. Right: the selected branch costate projected onto the normal and the fitted normal gradients. The reference changes branch near $s = 0$, producing a kink in the value and a jump in its normal derivative.

retains every row of the first set and adds 2,000 distinct augmentation samples. These shares count padding and collar samples selected within distance 0.5 of S ; the actual fractions inside the narrower evaluation tube $S_{0.3}$ are 15.0%, 23.3%, 32.4%, and 23.3%, respectively. All models use the same disjoint evaluation set of 994,558 points, including 91,104 inside $S_{0.3}$.

For the tested sets, increasing the augmentation share from approximately 23% to 40% or 60% does not reduce the error on $S_{0.3}$ (Table 4). The fully nested addition of 2,000 samples changes the ReLU^2 error from 0.262 to 0.263, and the Gaussian error from 0.303 to 0.321. Thus adding or reallocating samples near the switching set does not by itself remove the approximation difficulty in these single-seed fits.

Model	Training set	$S_{0.3}$	outside $S_{0.3}$
Gaussian	6k, 23% augmented	3.03×10^{-1}	2.12×10^{-1}
	6k, 40% augmented	3.15×10^{-1}	1.98×10^{-1}
	6k, 60% augmented	3.38×10^{-1}	2.61×10^{-1}
	6k + 2k augmented	3.21×10^{-1}	2.48×10^{-1}
ReLU ²	6k, 23% augmented	2.62×10^{-1}	1.55×10^{-1}
	6k, 40% augmented	2.82×10^{-1}	1.87×10^{-1}
	6k, 60% augmented	3.34×10^{-1}	2.85×10^{-1}
	6k + 2k augmented	2.63×10^{-1}	1.75×10^{-1}

TABLE 4. Relative H^1 error on evaluation samples inside and outside $S_{0.3}$. For each model and training set, the table reports the result with the smallest error on $S_{0.3}$ among the three tested values of α , with every entry in that row taken from the same run. The Gaussian uses $(\gamma, p) = (1, 2.01)$ and $\alpha \in \{10^{-3}, 10^{-4}, 10^{-5}\}$; ReLU² uses ψ_2 and $\alpha \in \{10^{-4}, 10^{-5}, 10^{-6}\}$. Errors are computed after un-normalizing, so the two blocks share one definition.

6.2.2. *Validation and regional errors.* Figure 14 tracks the best observed relative H^1 validation error at each support budget, for each activation paired with its penalty. The three nonhomogeneous activations use the parameter values in Figure 1b; the rectified powers use ψ_k with $\alpha = 10^{-6}$ for both $k = 2$ and $k = 3$. The smallest observed validation errors are 0.208 for ReLU², 0.209 for the Gaussian, 0.246 for ReLU³, 0.290 for tanh, and 0.469 for softplus.

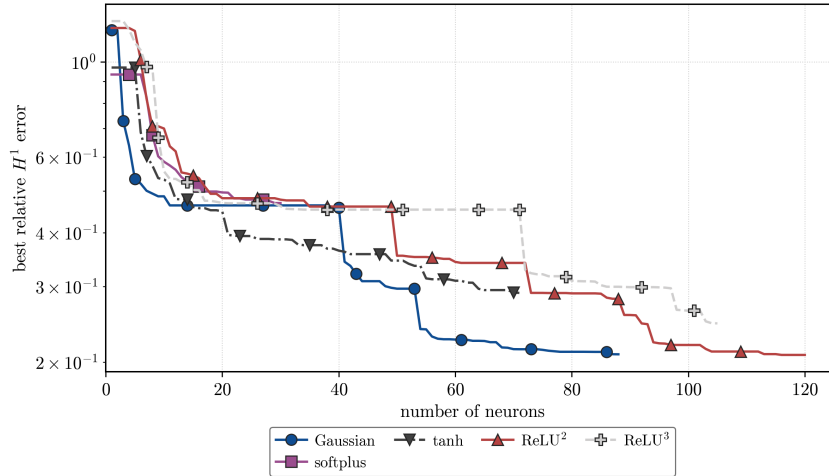


FIGURE 14. Best relative H^1 validation error against neuron count. The non-homogeneous activations use (56); the homogeneous activations use $J_{\psi_k}^M$ with $\alpha = 10^{-6}$ for both $k = 2$ and $k = 3$.

Figure 15 reports the relative H^1 error on a shared evaluation set of 994,558 points. The union of training rows across the production and oversampling datasets is excluded, and for each run we use the recorded iterate with the smallest training objective. Errors are reported on the subsets inside and outside $S_{0.3}$. The ReLU² fit has the smallest errors in both regions (0.275 and 0.155). The corresponding values are 0.290 and 0.175 for the Gaussian, 0.331 and 0.194 for ReLU³, 0.360

and 0.222 for tanh, and 0.521 and 0.511 for softplus. Every selected model has a larger relative error inside the switching tube than outside it.

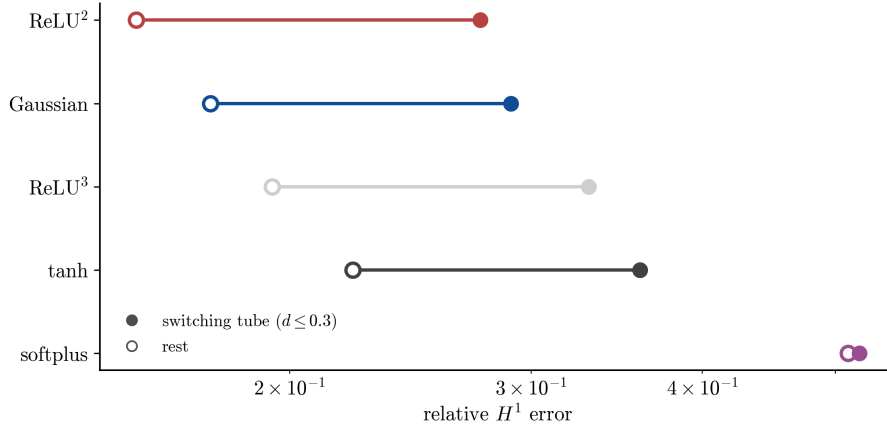


FIGURE 15. Relative H^1 error on $S_{0.3}$ (filled markers) and the remaining evaluation points (open markers), with rows ordered by the latter error.

6.2.3. *Feedback synthesis across the switching set.* Minimizing the Hamiltonian in u gives the optimal control in feedback form,

$$u(x) = -\frac{1}{2\eta m\ell^2} \partial_{\theta} V(x);$$

replacing V by a fitted $\hat{V}$ yields a candidate feedback law. We test each law in closed loop from two initial states placed on either side of the switching curve, $A = (0.74, 0.61)$ and $B = (0.26, 0.46)$, integrating the dynamics over the horizon $T = 10$ with fixed-step RK4 ($\Delta t = 5 \times 10^{-3}$), zero-order-hold controls, and the numerical bound $|u| \leq 30$ (the OCP itself is unconstrained). The sampled-PMP reference feedback searches the 40 nearest raw samples, including their shifts by $-4\pi, -2\pi, 0, 2\pi, 4\pi$, and uses the costate of the neighbour with the smallest stored value. This nearest-sample rule is a feedback heuristic; the separate open-loop comparison below solves the PMP boundary-value problem from each start.

From the harder start A , only ReLU^2 among the five nonconvex fits reaches the upright neighbourhood, at RK4-stage cost 85.3. The $\text{ReLU}-\ell^1$ baseline also reaches it, at cost 87.8, compared with 26.2 for the sampled-PMP reference (Table 5). Reaching the neighbourhood does not imply exact stabilization or near-optimal cost: the ReLU^2 trajectory settles near $\theta = 6.239$, retaining a nonzero running cost. Figure 17 shows the control signals from A . Each fitted law initially applies a larger negative control than the sampled-PMP reference, and the trajectories subsequently separate.

Figure 18 compares all five learned controllers with numerical open-loop references from both starts. As in the Van der Pol comparison, the reference for each start is the least-cost converged finite-horizon PMP solve initialized from the learned and dataset-feedback rollouts. It uses $T = 10$, a free terminal state, and zero terminal cost. This compares the realized truncated running costs directly; it does not estimate an infinite-horizon tail cost or certify global optimality of the reference.

The Gaussian, softplus, and tanh networks have 88, 30, and 70 neurons, respectively, with $(\alpha, \gamma, p) = (10^{-4}, 10, 2.01)$. The selected ReLU^2 and ReLU^3 fits have 120 and 105 neurons and both use $\alpha = 10^{-6}$, selected by switching-region error. For these checkpoints, only ReLU^2 reaches the stated upright neighbourhood from A ; all five do from B . These are the same

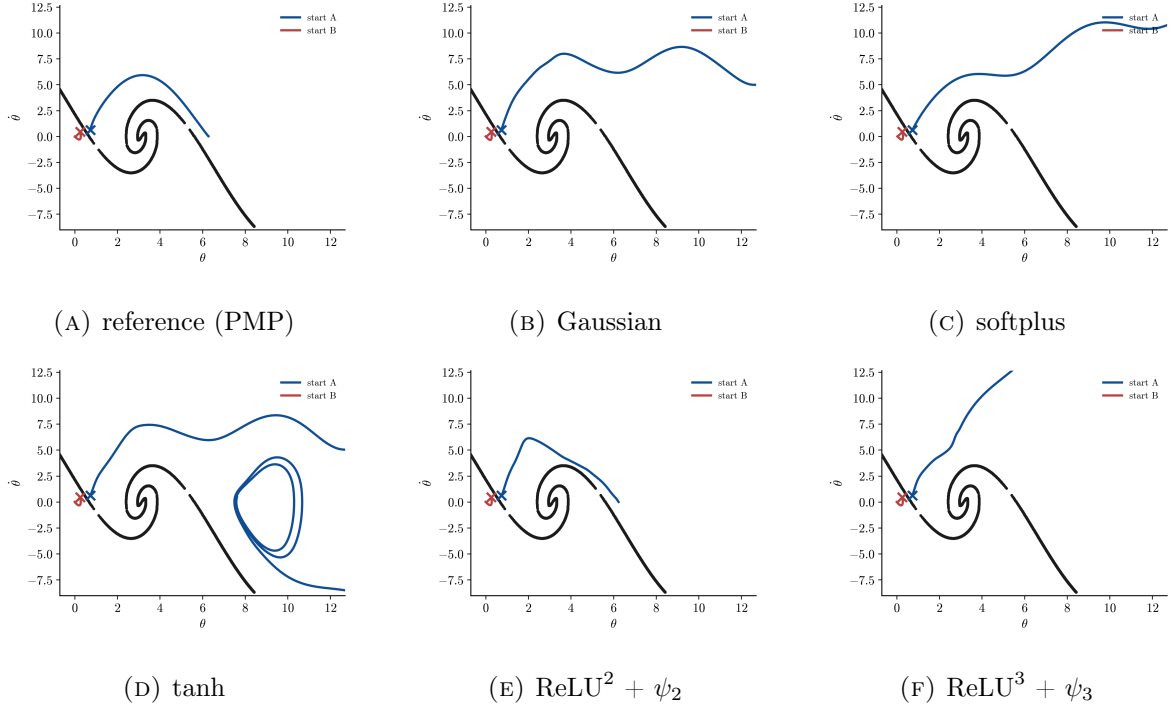
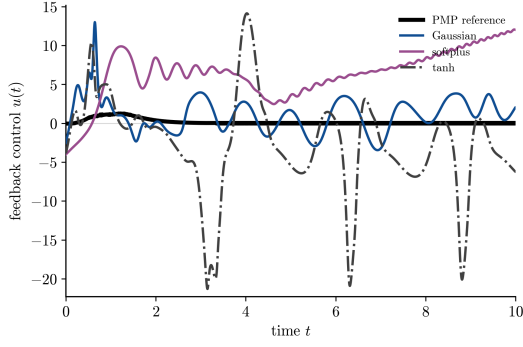


FIGURE 16. Closed-loop paths in the $(\theta, \dot{\theta})$ plane from the two starts A (blue) and B (red) on either side of the switching curve (black), one panel per feedback law; all panels share the same axes. The reference law swings over the top from A and brakes to $\theta = 0$ from B . Only ReLU^2 among these five learned laws reaches the stated upright neighbourhood from both starts; the other four do so only from B . Trajectories that leave the common plotting window continue to be integrated for the reported costs and terminal tests.

Feedback law	cost	upright
reference (PMP)	26.2	yes
$\text{ReLU} + \ell^1$	87.8	yes
Gaussian	361.4	no
softplus	6999.6	no
tanh	708.3	no
$\text{ReLU}^2 + \psi_2$	85.3	yes
$\text{ReLU}^3 + \psi_3$	69306.3	no

TABLE 5. Pendulum feedback from start $A = (0.74, 0.61)$ over $T = 10$. Costs use the held-control RK4 stages. The upright criterion is $\text{dist}(\theta, 2\pi\mathbb{Z}) < 0.4$ and $|\dot{\theta}| < 0.4$ at the terminal time. The reference is the sampled-PMP feedback heuristic. The nonhomogeneous models use $(\alpha, \gamma, p) = (10^{-4}, 10, 2.01)$. The $\text{ReLU}-\ell^1$ baseline uses $\alpha = 10^{-5}$, selected by validation error among $\alpha \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$. The rectified-power fits use $\alpha = 10^{-6}$, selected by switching-region error. Each law uses its run’s minimum-training-objective checkpoint.



(A) Algorithm 1.

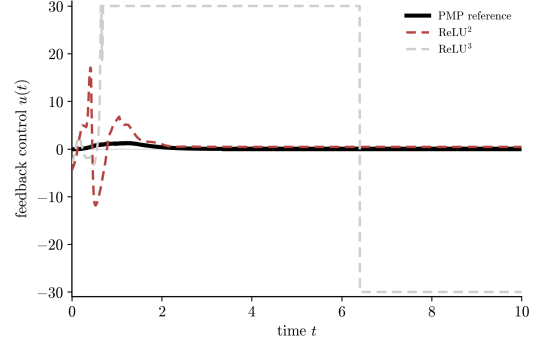
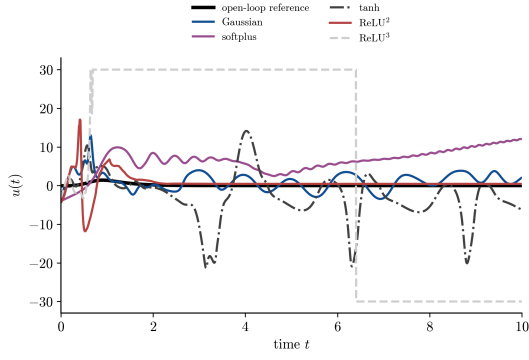
(B) ReLU^k models.

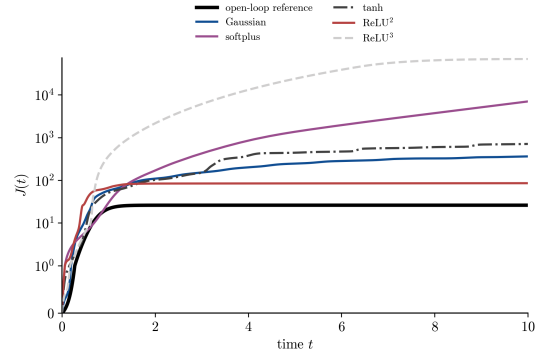
FIGURE 17. Feedback control $u(t)$ from start A , per law, beside the reference (black). The reference initially applies almost no control and selects the 2π swing-up branch. Every fitted law instead starts with a much larger negative control. After the trajectories separate, values at the same time are evaluated at different states and are therefore not pointwise feedback errors at a common state.

checkpoints used in the surfaces, regional comparisons, and feedback table. With the RK4-stage cost integration used in these panels, the costs from A are 361.4, 6999.6, 708.3, 85.3, and 69306.3 for the Gaussian, softplus, tanh, ReLU^2 , and ReLU^3 , respectively, versus 26.02 for the numerical open-loop reference. From B the learned costs range from 10.12 to 11.57, versus 10.10 for the reference. The large costs from A reflect trajectories that fail the terminal criterion, despite the numerical control guard.

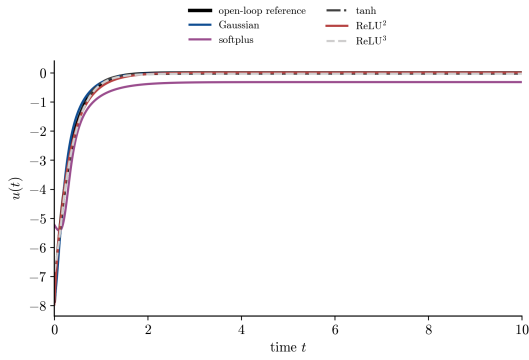
The ReLU^2 fit has the smallest reported regional errors and reaches the upright neighbourhood from both starts. Nevertheless, approximation accuracy alone does not guarantee the terminal outcome: the Gaussian has smaller global validation error than the $\text{ReLU}-\ell^1$ baseline (0.209 versus 0.300), yet only the latter reaches the upright neighbourhood from A . All five selected nonconvex laws reach it from B . Feedback synthesis near a switching set therefore requires a direct closed-loop assessment in addition to value and gradient errors.



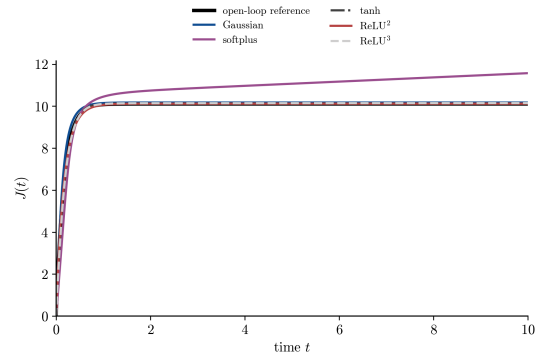
(A) Control from A.



(B) Accumulated cost from A.



(C) Control from B.



(D) Accumulated cost from B.

FIGURE 18. Pendulum controls and realized costs over $T = 10$. Each learned feedback law is simulated on its own trajectory with $\Delta t = 0.005$ and the numerical control bound $|u| \leq 30$. The black curves are numerical open-loop references with matching initial states and horizons. Costs $J(t) = \int_0^t \ell(x(s), u(s)) ds$ use the held-control RK4 stages for learned laws and three-point Gauss quadrature on the dense PMP solutions for the references. The feedback table uses the same RK4-stage costs. No predicted value or terminal-value estimate enters the new cost curves. For start A, the cost axis is linear below $J = 1$ and logarithmic above it to display the large differences between the controllers.

7. CONCLUSION AND OUTLOOK

We developed a measure-theoretic framework for sparse, gradient-augmented recovery of HJB value functions from PMP data. Its main contribution is a variational route from infinitely wide dictionaries to finite shallow networks for two activation classes that require different compactness arguments. For nonhomogeneous activations, simultaneous normalization of the measure and dictionary controls escape on the unbounded parameter domain and, together with the concave penalty, yields existence, finite support, separation and width bounds, and a quantitative decay rate for the excess above the insertion threshold. For positively k -homogeneous activations, radial optimization instead reduces the problem to the unit sphere and induces a fractional-power penalty; the resulting formulation likewise admits global minimizers and explicit finite-width conclusions. These results give a rigorous foundation for constructing sparse networks that approximate both a value function and its gradient.

The numerical studies illustrate the role and the boundary of this framework. On the Van der Pol problem, the learned feedback trajectories reach a small terminal neighbourhood of the origin. On the pendulum problem, all five selected laws reach the upright neighbourhood from the easier start. Among these five laws, only ReLU² does so from the harder start; the traditional ReLU- ℓ^1 baseline also succeeds there. The Gaussian has lower validation error than that baseline but fails the same terminal test, so approximation accuracy alone does not determine the closed-loop outcome. Thus gradient-augmented fitting is a useful bridge from PMP data to feedback synthesis, while feedback performance must still be assessed directly when the value gradient is discontinuous.

The most immediate extension concerns the insertion step: the rate analysis assumes exact global candidate maximization, whereas the implementation uses a multistart local search. Developing certified or more reliable candidate searches would therefore bring the algorithm closer to the theoretical guarantee. A second direction is to complement the present approximation criteria with feedback-aware training or validation near switching sets, and then to examine the resulting methods in higher-dimensional control problems.

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